Introduction to simplicial sets

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Abstract

We introduce the theory of simplicial sets and some basic simplicial homotopy theory. We try to make this paper accessible to those with a grasp of basic category theory and point-set topology, and not necessarily algebraic topology. For this reason, although we acknowledge the geometric intuition and some topological aspects of the theory, we avoid topology when possible and prefer to work in a more categorical setting.

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1 Introduction

Simplicial sets have been used extensively in algebraic topology as a way to encode complex information about topological spaces into a *relatively* simple combinatorial object, which often allows for more streamlined study of aspects of the space. More recently, they've become extremely useful in category theory, especially homotopy theory and higher category theory, due to their ability to encode homotopic information. A well known example of this is quasi-categories, a certain type of simplicial set which forms a convenient model for $(\infty, 1)$-categories, which have been studied extensively by André Joyal in [Joy08a] and [Joy08b], and Jacob Lurie in [Lur09].

Unfortunately, simplicial sets are just complicated enough to be difficult for beginners to work through easily, but are just simple enough for experts to consider them fairly trivial objects, and as such there are very few expositions on simplicial sets which are readily understandable by the amateur. For example, the references [GJ99] and [May67] are considered standard on the topic, but the former is quite advanced for a first pass, and the latter is described by this author as being "the densest textbook in existence."

Friedman recently wrote [Fri23] which we believe to be a very good introduction, but it focuses much more on the topological aspects of the theory than we would like. We also like Emily Riehl's [Rie11] which is very categoricaly motivated, but we feel her introduction strays too far from the geometric roots of the subject for a first introduction. Given the prominence of simplicial sets in category theory, we would like for there to be an entry level resource that focuses on the categorical motivations of the theory, without fully forsaking the incredible intuition lended to the subject by geometry. This paper is our attempt to produce such a resource.

Category theory is the main prerequisite of this paper. Any student with an understanding of categories, limits, adjunctions, and an undergrad understanding of point-set topology should be more than equipped to read this paper. We recommend [Rie16] and [Lei14] as good resources to learn category theory from, and [ML98] as an excellent reference, although we discourage learning from here.

Throughout the paper, we work in the standard category $\mathcal{T}op$ whenever possible, and only pass to the category $\mathcal{CG}$ of compactly generated spaces when needed. We do this to avoid overloading readers who may not have had advanced topology exposure, especially since passing to $\mathcal{CG}$ is a largely technical point. Most of the theory developed herein works the same in $\mathcal{T}op$ as in $\mathcal{CG}$ up to homotopy equivalence.

2 Geometric simplices

2.1 The standard n-simplices

The standard n-simplex is a fancy word for "an n-dimensional equilateral triangle." If you are reading this, you most likely know what a regular, 2-dimensional equilateral triangle is, so let's begin by defining the standard 2-simplex. This involves promoting our intuitive understanding of a triangle to a proper, point-set definition.

There are a number of ways of doing this, but the standard way is as follows. We define the standard 2-simplex Δ^2 as the set

$$\Delta^2 := \left\{ (t_0, t_1, t_2) \in \mathbb{R}^3 \mid \sum_{i=0}^{2} t_i = 1, \text{ and } t_i \geq 0 \text{ for all } i \right\} \subseteq \mathbb{R}^3.$$

Let's parse this definition. We first note that each coordinate t_i must be non-negative. We also note that $t_0 + t_1 + t_2 = 1$. If we graph $x + y + z = 1$ in $\mathbb{R}^3$, and restrict to only the first octant, as shown in fig. 1, then the resulting shape is an equilateral triangle. Of course, there are other ways to define a triangle in $\mathbb{R}^3$, or even $\mathbb{R}^2$, but this is what the mathematical community has decided to choose as the "standard way", since this construction simplifies some later constructions such as the maps of definition 2.4.1.

One of the reasons we choose to define the standard 2-simplex as this specific construction is because it is fairly clear how to generalize it. The ordered triple (t_0, t_1, t_2) is indexed from 0 to 2, so a standard n-simplex should likely involve an n-tuple indexed from 0 to n, of the form $(t_0, \ldots, t_n)$. We should still require $t_i \geq 0$ for all i. Finally, since we require the equality $t_0 + t_1 + t_2 = 1$, we can just generalize this to the sum over all entries $t_0 + \cdots + t_n = 1$. To this end, we offer the fully general defintition:

Definition 2.1.1. The *standard n-simplex*, denoted by Δ^n, is the set

$$\Delta^n := \left\{ (t_0, \ldots, t_n) \in \mathbb{R}^{n+1} \mid \sum_{i=0}^{n} t_i = 1 \text{ and for all } i,\ t_i \geq 0 \right\} \subseteq \mathbb{R}^{n+1}.$$

♡

Terminology 2.1.2. The plural form of the word "simplex" is "simplices". The text that this author first learned simplicial objects from did not clarify this, and this author was very confused by this fact. Note

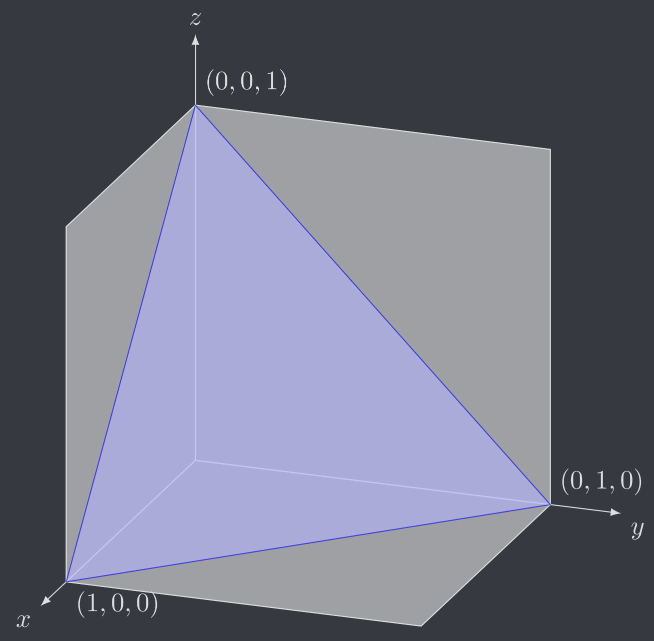

Figure 1: The standard 2-simplex Δ^2

that the terms "simplex" and "simplices" inherit a new but similar definition in the context of simplicial sets, which we make precise in terminology 3.1.6. However, "simplices" will still remain the plural of "simplex" in the new defintition.

Remark 2.1.3. The reader who is familiar with topology will frequently see us refer to n-simplices as topological spaces. The obvious choice for a topology on Δ^n is the subspace topology inherited from $\mathbb{R}^{n+1}$, and we will always leave this definition implicit. The reader who is *not* familiar with topology should be able to safely ignore this technicality, as we won't often have a need to work with the topological properties of spaces in this paper. However, we caution the reader not to go further down the rabbit hole without having learned at least some basic topology.

Example 2.1.4. We describe the standard 0- and 1-simplicies.

1. The standard 0-simplex is the subspace $\{t_0 \in \mathbb{R} \mid t_0 = 1\} = \{1\}$. In particular, Δ^0 is a singleton.
2. The standard 1-simplex is the set of all pairs $(t_0, t_1) \in \mathbb{R}^2$ such that $t_0, t_1 \geq 0$ and $t_0 + t_1 = 1$. Rearranging this equation gives $t_1 = 1 - t_0$. Graphing this equation gives fig. 2, which depicts the standard 1-simplex as a simple line segment.

Remark 2.1.5. An n-simplex is "standard" if it's point-set definition is precisely the same as definition 2.1.1. If a space has the same "shape"[1] of the standard n-simplex, but has a different point-set definition (for example, defined in $\mathbb{R}^{n+2}$, or translated to a different location, or rotated), then we simply call that an n-simplex. With this in mind, we make a series of observations based on figs. 1 and 2.

1. Note that the vertices of both Δ^1 and Δ^2 are just points, but we also know that Δ^0 is just a point. Thus, vertices are simply 0-simplicies.
2. Note that the edges of Δ^2 are line segments, and in particular they are the same line segment as the standard 1-simplex of fig. 2, becuase they trace a diagonal path through a unit square. Thus, edges are simply 1-simplicies. Also note that edges are always inserted *between two vertices*, or alternatively in view of (1), *between two 0-simplicies.*
3. At this point, we likely intuitively understand that the standard 3-simplex is a tetrahedron: a 4-sided

[1] The notion of "shape" here is made precise by defining n-simplices as the convex spanned by a set of n affinely independent points in $\mathbb{R}^m$, but all of our n-simplices are constructed in such a way that this rigorous definition is not necessary whatsoever, so the intuitive definition suffices for our paper.

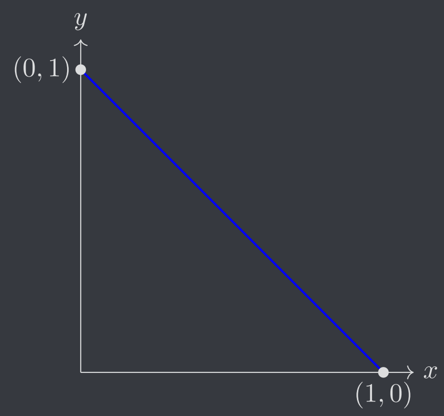

Figure 2: The standard 1-simplex

triangular pyramid, wherein all sides are equilaterial triangles. In particular, the sides of the 3-simplex are 2-simplicies, which are inserted between three edges (1-simplicies), which are inserted between two vertices (0-simplicies).

4. Finally, note that the standard 0-simplex has one vertex, the standard 1-simplex has two vertices, the standard 2-simplex has three vertices, and the standard 3-simplex has four vertices.

Inded, these remarks generalize to n-simplicies, meaning we can gain the following intuition. To construct the standard n-simplex, we can start with $n+1$ vertices, draw edges between each two vertices, draw 2-simplicies between each three edges, draw 3-simplicies between each four 2-simplicies, and so on. Oftentimes we will discuss the k-simplices of an n-simplex. This simply means the k-simplices inserted in this "construction process".

Terminology 2.1.6. In view of remark 2.1.5, we introduce some standard terminology for simplices. A *face* of the n-simplex is an $(n-1)$-simplex of the n-simplex. The vertices of the *standard* n-simplex can be precisely described as points of the form $(0,\ldots,1,\ldots,0)$; that is, tuples where there is only one nonzero entry, which must be 1 by definition 2.1.1. Vertices of the *standard* n-simplex thus inherit a canonical ordering: the ith vertex of Δ^n is the vertex where the 1 appears in the ith spot. For example, the 1st vertex of Δ^2 is $(1,0,0)$, the second is $(0,1,0)$, and the third is $(0,0,1)$. This also provides a canonical ordering for faces. Each face is exactly across from a vertex[2]. We call the face directly across from the ith vertex the *ith face*. n-simplicies which are "nonstandard" do not inherit a *canonical* ordering, but we can still choose an arbitrary ordering of the vertices, and an ordering of the faces follows by the same logic.

Example 2.1.7. The faces of the standard 2-simplex are precisely it's edges. In figure 3, we label the faces and the vertices of the standard 2-simplex by the canonical ordering of terminology 2.1.6. The ith vertex is laveled v_i, and the ith face is labeled f_i.

Exercise 2.1.8. Unfortunately we cannot draw the *standard* 3-simplex, since it lives in 4 dimensions, so draw any 3-simplex, and give its vertices an arbitrary ordering. Label all of its faces.

2.2 The category $\mathbf{\Delta}$

The category $\mathbf{\Delta}$ encodes valuable combinatorial data about the standard n-simplicies, which will be exploited heavily to develop the theory of simplicial sets in §3. Unfortunately, seeing the usefulness of this layer of abstraction requires the reader to wade through some poorly motivated material, but we hope the reader will see the motivation by the end. We begin by reviewing the definition of (finite) ordinal numbers.

Definition 2.2.1. The first ordinal number is defined as the empty set $\varnothing$, and is denoted 0. The $(n+1)$th ordinal number is defined inductively as $n \cup \{n\}$. ♡

[2] To rigorize this notion, construct a perpendicular line through the middle of a face and see which vertex it passes through.

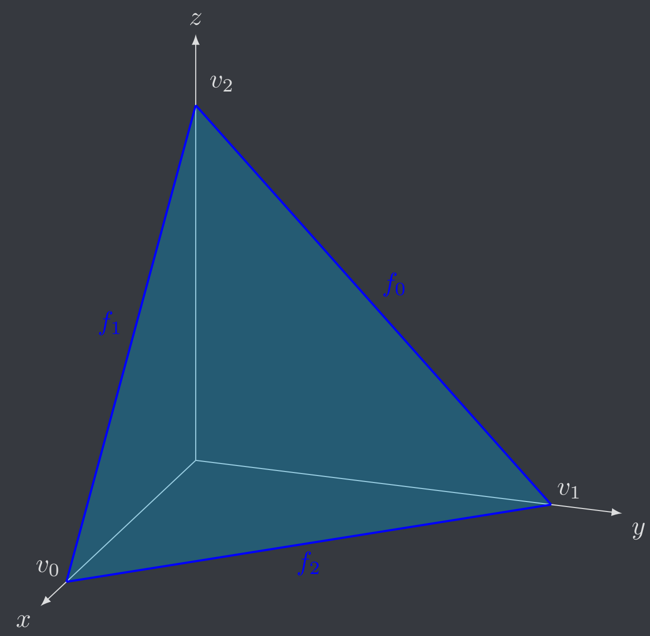

Figure 3: The standard 2-simplex, labeled

Exploring this defintion, we see that $0 = \{\}$, and $1 = 0 \cup \{0\} = \{\} \cup \{0\} = \{0\}$. We then see

$$2 = 1 \cup \{1\} = \{0\} \cup \{1\} = \{0, 1\}, \qquad 3 = 2 \cup \{2\} = \{0, 1\} \cup \{2\} = \{0, 1, 2\}.$$

In general, we might think that $n = \{0, \ldots, n-1\}$, and indeed this is readily proven for the finite case:

Proposition 2.2.2. *If n is a nonzero finite ordinal number, then n is defined as $\{0, \ldots, n-1\}$.*

Proof. We use the method of proof by induction. By our above inspection, $1 = \{0\} = \{0, \ldots, 0\}$. We assume the inductive hypothesis that $n = \{0, \ldots, n-1\}$. Then by definition 2.2.1,

$$n + 1 = n \cup \{n\} = \{0, \ldots, n-1\} \cup \{n\} = \{0, \ldots, n\}.$$

By induction, the statement is proven. ■

Notation 2.2.3. Category theory rarely deals with numbers in the conventional sense of the term, so when we say "consider the finite ordinal n", we are likely using this as a shorthand for the set $\{0, \ldots, n-1\}$. It is also often convenient to havea second notation for finite ordinals: $[n]$ denotes the set $n + 1 = \{0, \ldots, n\}$. We then write $[-1]$ for $0 = \{\}$. In other words, we are denoting a linearly ordered set $\{0, \ldots, n\}$ by its greatest element.

Remark 2.2.4. In view of notation 2.2.3, it is important to note that the notation $[n]$ comes up almost exclusively in the context of simplicial objects. Note that these sets are *ordered*, since the relation $\leq$ defines an order on the set. Recalling the definition of poset categories, one often will say "consider the category $\mathbf{n}$". What they really mean is "consider the category whose objects are elements of n, and whose morphisms are defined by taking the poset structure on n using the $\leq$ relation." Most authors use bold font for ordinals to denote when they are being considered as categories. Although we dislike this convention, we agree it is pedagogically beneficial, so we adopt it for this paper.

Notation 2.2.5. We write ω for the first infinite ordinal. As a set, ω contains all finite ordinals $\{0, 1, 2, \ldots\}$. The natural numbers $\mathbb{N}$ can in fact be defined as the number ω.

Definition 2.2.6. The *simplex category* $\mathbf{\Delta}$ is the category where

1. Objects are nonempty finite ordinals, meaning $[n]$ for $n \geq 0$;

2. Morphisms are weakly order preserving functions, meaning a morphism is a function $f : [n] \to [m]$ such that $x \leq y$ implies $f(x) \leq f(y)$.

♡

It is not immediately clear how the simplex category relates to definition 2.1.1, but there is an important functor that will make the relationship more clear. For now, we will only discuss the action of this functor on objects, since we will need to further develop the theory of morphisms in $\mathbf{\Delta}$ before the action of the functor can make sense on arrows. We thus offer the following preliminary definition:

Definition 2.2.7. The functor $\Delta : \mathbf{\Delta} \to \mathcal{T}\mathrm{op}$ is defined by mapping the finite ordinal $[n]$ to the standard n-simplex Δ^n, as defined in definition 2.1.1. ♡

Notation 2.2.8. We have that $\Delta[n] = \Delta^n$, so we will often choose to write the functor of notation 2.2.5 as $\Delta^{\bullet}$. The big dot, called a *bullet*, is interpreted as a placeholder for the value of the functor, similar to the "$-$" in hom-functors.

Remark 2.2.9. The following remark is not important for this paper, but is an interesting fact. The category $\mathbf{\Delta}$ admits a full embedding into the category $\mathcal{C}\mathrm{at}$ of small categories. We map each ordinal $[n-1]$ to the poset category $\mathbf{n}$, and functors $F : \mathbf{n} \to \mathbf{m}$ are precisely order preserving functions $F : [n-1] \to [m-1]$.

2.3 Cofaces and codegeneracies

We now introduce an extremely important class of maps in the simplex category $\mathbf{\Delta}$, which in fact subsume all other morphisms in the category. These are coface and codegeneracy maps. We adhere to the notational conventions of [GJ99] for simplicial and cosimplicial operators.

Definition 2.3.1. Let $[n]$ be an object in $\mathbf{\Delta}$, and let $i \in [n]$. The *coface map* $d^i : [n-1] \to [n]$ is defined as

$$d^i(x) = \begin{cases} x, & x < i \\ x+1, & x \geq i \end{cases}.$$

Similarly, if $i \in [n]$, then the *codegeneracy map* $s^i : [n+1] \to [n]$ is defined as

$$s^i(x) = \begin{cases} x, & x \leq i \\ x-1, & x > i \end{cases}.$$

♡

Intuition 2.3.2. Coface and codegeneracy maps should be understood as follows. For now, disregard the naming of these maps; we will see why these maps are named the way they are once we consider geometric applications.

1. Coface maps insert $[n-1]$ into $[n]$ in a specific way. The ith coface map d^i skips the number i, so it inserts all the numbers exactly where they should be ($x \mapsto x$) *until it gets to* i. Once it gets there, it skips i and starts inserting the numbers one slot above where they should be ($x \mapsto x+1$). In particular, cofaces are injective.
2. Instead of skipping i, the ith codegeneracy *double counts* i. What it does is it once again maps all numbers to themselves ($x \mapsto x$) until it gets to i. It then maps both i and $i+1$ to i. Note that to do this, it had to map $i+1$ down a slot, so it continues mapping all the rest of the numbers down a slot ($x \mapsto x-1$). In particular, codegeneracies are surjective.

Exercise 2.3.3. One can intuitively understand cofaces and codegeneracies by explicitly drawing out all of the arrows. For example, the coface $d^2 : [2] \to [3]$ can be described by the diagram

$$\begin{tikzcd} & 3 \\ 2 \arrow[ur, mapsto] & 2 \\ 1 \arrow[r, mapsto] & 1 \\ 0 \arrow[r, mapsto] & 0 \end{tikzcd}$$

Draw out similar diagrams for $d^2 : [5] \to [6]$, $s^0 : [3] \to [2]$, and $s^2 : [4] \to [3]$.

Proposition 2.3.4. *Every morphism f in $\mathbf{\Delta}$ admits a unique representation as a composite of cofaces and codegeneracies. In particular, f can be written uniquely as a composition*

$$f = d^{i_1} \circ \cdots \circ d^{i_k} \circ s^{j_1} \circ \cdots \circ s^{j_h}.$$

Proof. We follow the proof of [ML98, VII.5], as recommended by [GJ99]. Induction on $i \in [n]$ shows that a function $f : [n] \to [m]$ is uniquely determined by its image in $[m]$, together with the set of those $j \in [n]$ such that $f(j) = f(j+1)$. We call these elements j "elements where f does not increase." Index the elements of $[m]$ not in the image of f by $i_1, \ldots, i_k$, and index the elements where f does not increase by $j_1, \ldots, j_h$. Put these in reverse order and form the composite indicated in the statement of the proposition. The equality follows. ■

Exercise 2.3.5. Complete the proof of proposition 2.3.4. Although the argument is correct, it is clearly not fully rigorous, as it claims nonobvious things follow without justification. For example, although the induction argument it references is not difficult, it is not fully obvious that it should be true without completing the argument.

Proposition 2.3.6 (Cosimplicial identities)**.** *With domains and codomains as appropriate, the following list of identities for the coface and codegeneracy operators hold, and are referred to as the* cosimplicial identities*.*

$$\begin{cases} d^j d^i = d^i d^{j-1}, & i < j \\ s^j d^i = d^i s^{j-1}, & i < j \\ s^j d^j = s^j d^{j+1} = 1 & \\ s^j d^i = d^{i-1} s^j, & i > j+1 \\ s^j s^i = s^i s^{j+1}, & i \leq j \end{cases}.$$

Exercise 2.3.7. It is a rite of passage to prove the cosimplicial identities of proposition 2.3.6. They may be regarded as a corollary of proposition 2.3.4 by inspecting the proof, but it is generally more simple and more fulfilling to do a direct proof. They are not very difficult; in fact you can likely see how to prove some of them in your head.

Remark 2.3.8. By proposition 2.3.4, cofaces and codegeneracies subsume all other morphisms in $\mathbf{\Delta}$. This is made even more precise by [ML98, VII.5 Prop. 2], which claims that by applying the cosimplicial relations of proposition 2.3.6, one can obtain any morphism in $\mathbf{\Delta}$ from cofaces and codegeneracies. This is typically phrased as the claim that cofaces and codegeneracies, subject to the cosimplicial relations, generate the maps in $\mathbf{\Delta}$.

2.4 The functor $\Delta^\bullet : \mathbf{\Delta} \to \mathcal{T}op$

We now return to our investigation of the functor $\Delta^\bullet$, defined on objects in definition 2.2.7. This investigation will not only provide us with a geometric intuition for the simplex category, cofaces, and codegeneracies, but will also offer a geometric intuition for the cosimplicial identities of proposition 2.3.6. We thank [May99] for being one of the best[3] resources we found which explicitly defined the maps of definition 2.4.1.

Definition 2.4.1. Let Δ^n denote the standard n-simplex, as in definition 2.1.1. For each $i \in [n]$, define a map $d^i : \Delta^{n-1} \to \Delta^n$ given by

$$(t_0, \ldots, t_{n-1}) \longmapsto (t_0, \ldots, t_{i-1}, 0, t_i, \ldots, t_{n-1}).$$

We call d^i the ith *coface map*. Similarly, for each $i \in [n]$, define a map $s_i : \Delta^{n+1} \to \Delta^n$ by

$$(t_0, \ldots, t_i, t_{i+1}, \ldots, t_{n+1}) \longmapsto (t_0, \ldots, t_i + t_{i+1}, \ldots, t_{n+1}).$$

We call s^i the *codegeneracy map*. ♡

[3]Translation: the only.

Remark 2.4.2. As I'm sure the reader can readily infer on their own, the definition of the functor $\Delta^\bullet$ on morphisms is given by mapping cofaces $d^i \mapsto d^i$, codegeneracies $s^i \mapsto s^i$, and all other maps are identified via their decomposition in proposition 2.3.4. We will explore what this means geometrically in intuition 2.4.4 after a short commentary on terminology.

Terminology 2.4.3. It is far more common to refer to the maps d^i and s^i of definition 2.4.1 as *face* maps and *degeneracy maps*, but as we will see in notation 3.1.2, this conflates with common terminology for simplicial sets. In particular, we prefer to use the 'co' prefix for covariant functors $\mathbf{\Delta} \to (-)$, and we drop it for contravariant functors $\mathbf{\Delta}^{\mathrm{op}} \to (-)$. However, our categorical interpretation originaly arose from the study of simplices in algebraic topology, and before the categorical framework was developed it would have obviously been odd to call face maps "coface maps". As such, the terminology for these maps has persisted in the special case of the functor $\Delta^\bullet : \mathbf{\Delta} \to \mathcal{T}\mathrm{op}$. Although this author prefers to adhere to the classical convention, we add in the 'co' prefix in this paper for pedagogical purposes, as we believe it may prevent confusion. It is important to note that **basically no other papers use this terminology**, and we use it *only* for pedagogical purposes.

Intuition 2.4.4. Now we explore the geometric intuition of the coface and codegeneracy maps between geometric simplices.

- (cofaces) The intuition for cofaces is fairly simple. Consider the coface $d^i : \Delta^{n-1} \to \Delta^n$. Recall by terminology 2.1.6 that the faces of the standard n-simplex can be ordered according to the canonical ordering of the vertices. In particular, the faces are ordered from 0 to n, or equivalently, they are ordered by the set $[n]$. Thus each $i \in [n]$ corresponds to a *face* of the standard n-simplex, which we recall by remark 2.1.5, is simply an $(n-1)$-simplex. The ith coface map takes the standard $(n-1)$-simplex, and inserts it as the ith face of the standard n-simplex.

  To see how this intuition arises from the definition in definition 2.4.1, we consider a simple example. Consider the coface $d^1 : \Delta^1 \to \Delta^2$. We map $(t_0, t_1) \mapsto (t_0, 0, t_1)$. This places the 1-simplex on the xz-plane, as illustrated in fig. 4.

  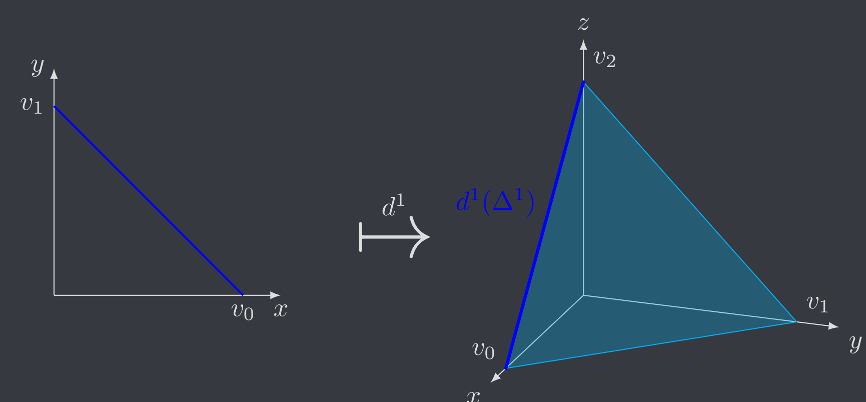

  Figure 4: Visualization of $d^1 : \Delta^1 \to \Delta^2$

  The idea is thus that the edges of the standard 2-simplex are precisely those that intersect with one of the xy-, yz-, or xz-planes. Since the map d^i keeps one coordinate 0, its image is necessarily one of these planes. And of course, this intuition generalizes to higher dimensions. Also, it should be fairly obvious why we identify the coface maps of $\mathbf{\Delta}$ with these coface maps between simplicies; setting one coordinate always equal to 0 is very similar to "skipping" the correspoding number, which gives us the intuitive link between the two notions.

- (codegeneracies) The intuition for codegeneracies is not as simple. However, after we see the intuition, we will not only see how these behave geometrically, but we will also see why this definition of codegeneracies makes sense with regards to the definition in $\mathbf{\Delta}$. In fact, we start with the intuition in $\mathbf{\Delta}$, and translate this to define a map between simplices. We then illustrate the fact that definition 2.4.1 agrees with the intuition we give here.

Recall that the codegeneracy $s^i : [n+1] \to [n]$ inserts both i and $i+1$ at i, so we can think of it as *identifying* i and $i+1$ with each other. The functor $\Delta^\bullet$ can also be thought of as identifying each $i \in [n]$ with an entry t_i in an n-tuple. So if s^i identifies i and $i+1$ together, and if $\Delta^\bullet$ identifies these numbers with entries in a tuple, we can think of the induced map as *identifying* the ith and $(i+1)$th entries in the tuple. In other words, we are collapsing the entire $i,(i+1)$-plane generated by these entries into a single line, generated by the resultant single entry. We visualize this in fig. 5.

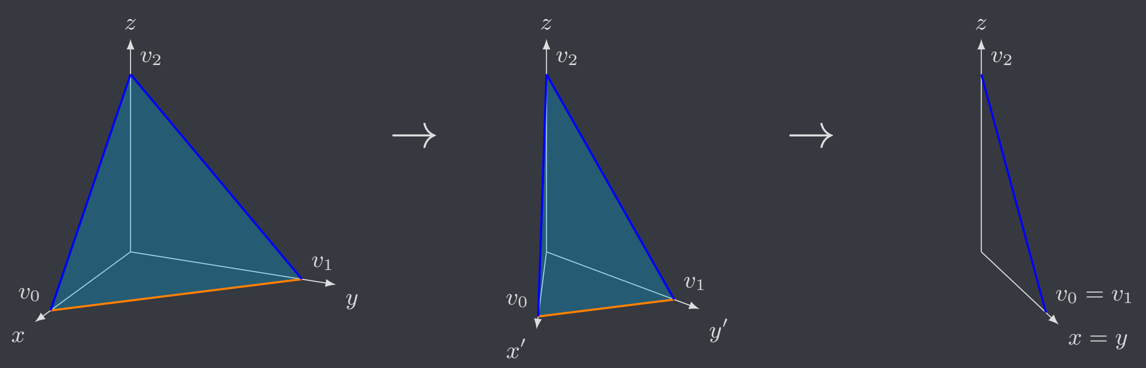

Figure 5: Visualization of $s^0 : \Delta^2 \to \Delta^1$

We call simplicies like the one in fig. 5 *degenerate*. Degenerate simplicies are n-simplicies where two of the vertices coencide, meaning the n-simplex can be regarded as an $(n-1)$-simplex. As seen in fig. 5, the vertices v_0, v_1 are identified under s^0, allowing us to regard $s^0(\Delta^2)$ as a 1-simplex. However, it can still be viewed as a 2-simplex with two vertices that coencide, making the simplex *degenerate*. Hence the name of the codegeneracy maps.

Although this description may make intuitive sense, it is nontrivial to connect it back to definition 2.4.1. The easiest way to see that these notions are equivalent is to consider the action on the vertices. To see that the ith and $(i+1)$th vertices are indeed identified, recall that the ith vertex is the point $(0_0, \ldots, 1_i, \ldots, 0_{n+1})$, and the $(i+1)$th vertex is the point $(0_0, \ldots, 0_i, 1_{i+1}, \ldots, 0_{n+1})$, where we have denoted the ordering of each entry with a subscript. We then obtain

$$s^i(0_0, \ldots, 1_i, 0_{i+1}, \ldots, 0_{n+1}) = (0_0, \ldots, 1_i + 0_{i+1}, \ldots, 0_{n+1})$$

$$= (0_0, \ldots, 0_i + 1_{i+1}, \ldots, 0_{n+1}) = s^0(0_0, \ldots, 0_i, 1_{i+1}, \ldots, 0_{n+1}).$$

This is the simplest way to connect the intuition herein back to definition 2.4.1. Since addition in the manner of definition 2.4.1 is well known to be continuous, attempting to imagine a continuous map that identifies the respective vertices will generally lead back to the intuition we gave here. We encourage the reader who finds this unconvincing to work out a simple toy example for some $s^i : \Delta^2 \to \Delta^1$ or $s^0 : \Delta^1 \to \Delta^0$.

Remark 2.4.5. Intuition 2.4.4 allows for a rigorous definition of the ith face of the standard n-simplex as the image of the ith coface $d^i : \Delta^{n-1} \to \Delta^n$.

The functor $\Delta^\bullet : \mathbf{\Delta} \to \mathcal{T}\mathrm{op}$, as far as we are aware, does not have a name, and is simply referred to as the "canonical functor $\mathbf{\Delta} \to \mathcal{T}\mathrm{op}$". However, this functor is of incredible importance, for reasons the reader is likely already realizing. We illuminated in remark 2.1.5 how n-simplices are constructed from lower level simplicies, and the *coface* maps of definition 2.4.1 express the same information for the standard n-simplices. However, we have also obtained *codegeneracy* maps, which are ways of turning higher level simplices into lower level simplices. The relationships between geometric simplices are thus encoded in their cofaces and codegeneracies, which the functor $\Delta^\bullet$ shows are also encoded in the category $\mathbf{\Delta}$.

The conclusion of this section is that $\mathbf{\Delta}$ offers us a combinatorial model for studying the relationships of the simplices. For example, consider the cosimplicial identities as stated in proposition 2.3.6. One of these identities states that for $i < j$, $d^j d^i = d^i d^{j=1}$. We can use the functor $\Delta^\bullet$ to give geometric meaning to this statement by interpreting it as a relationship between cofaces of simplices. Geometrically, this means that inserting the $(n-2)$-simplex into the ith face of the $(n-1)$-simplex, and then inserting that into the jth face of the n-simplex, is the same as inserting the $(n-2)$-simplex into the $(j-1)$th face of the $(n-1)$-simplex, and then inserting that into the ith face of the n-simplex. We see that this complex statement is simplified considerably by considering the combinatorial data of the category $\mathbf{\Delta}$, rather than the geometric data given by simplices. The other cosimplicial identities tell us similar statements:

- Let $i < j$. Inserting the n-simplex into the ith face of the $(n+1)$-simplex, and then identifying the jth and $(j+1)$th vertices of the $(n+1)$-simplex to produce an n-simplex, is the same as identifying the $(j-1)$th and jth vertices of the n-simplex to produce an $(n-1)$-simplex, then inserting that simplex into the ith face of the n-simplex.
- Inserting the n-simplex into *either* the jth face *or* the $(j+1)$th face of the $(n+1)$-simplex, and then identifying the jth and $(j+1)$th vertices to produce an n-simplex, is the same as doing nothing.
- Let $i > j+1$. Inserting the n-simplex into the ith face of the $(n+1)$-simplex, and then identifying the jth and $(j+1)$th vertices to form an n-simplex, is the same as identifying the jth and $(j+1)$th of the n-simplex to form an $(n-1)$-simplex, and then inserting it into the $(i-1)$th face of the n-simplex.
- Let $i \leq j$. Identifying the ith and $(i+1)$th vertices of the $(n+2)$-simplex to form an $(n+1)$-simplex, and then identifying the jth and $(j+1)$th vertices to form an n-simplex, is the same as identifying the $(j+1)$th and $(j+2)$th vertices of the $(n+2)$-simplex to form an $(n+1)$-simplex, and then identifying the ith and $(i+1)$th vertices to form an n-simplex.

Clearly, many of these claims are much easier proven as the cosimplicial identities in $\mathbf{\Delta}$, rather than directly proving them as properties of the geometric cofaces and codegeneracies. This understanding shows us that the simplex category $\mathbf{\Delta}$ is not only closely related to the geometric simplices, but is also an extremely neat and compact way of storing complex combinatorial data, allowing proofs to be simplified considerably. In the next section, we use this understanding to define *simplicial sets*, which are similar and more powerful ways of storing combinatorial data. Simplicial sets are related to geometric simplices via the *geometric realization functor*, an extremely important concept which will be introduced in section 3.2.

3 Simplicial sets

So far, we have learned about geometric simplices, their cofaces and codegeneracies, and how the simplex category $\mathbf{\Delta}$ encodes this information combinatorially. As promised, we now explore simplicial sets as a more powerful way to encode combinatorial data about various structures. Like the simplex category, simplicial sets are closely related to topological spaces and the geometric simplicies because they admit a *geometric realization*, which is a process for constructing topological spaces out of simplicial sets. The use of simplicial sets is ubiquetous throughout algebraic topology and homotopy theory, and we will see some applications in section 4.

In this section, we will have to consider universal constructions such as quotients and coproducts in the category $\mathcal{T}\mathrm{op}$. For the topologically uninitiated reader, we make the following remark:

Remark 3.0.1. There is a forgetful functor $U : \mathcal{T}\mathrm{op} \to \mathcal{S}\mathrm{et}$ which admits both a left *and* a right adjoint. In particular, U preserves all limits and colimits. Since U is a forgetful functor, it does not change the underlying set of a topological space, and thus all universal constructions in $\mathcal{T}\mathrm{op}$ have the same underlying set as those in $\mathcal{S}\mathrm{et}$.

3.1 Definition

The definition of simplicial sets is fairly innocuous, but the level of detail that our problems demand we work in will reveal the definition to be suprisingly complex. Most experts find simplicial sets very simple, but students often take significant time to fully process the definition and to come to terms with it. We develop many examples and exercises throughout this section to improve the reader's understanding. We highly recommend to work through all the exercises.

Definition 3.1.1. A simplicial set $X_\bullet$ is a contravariant functor $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$. We generally denote the image $X[n]$ on an object by the subscript X_n, hence the bullet notation. We will generally not denote the action on morphisms by subscripts, and instead appeal to notation 3.1.2. ♡

Notation 3.1.2. By proposition 2.3.4, we need only consider the action of a simplicial set X on cofaces and codegeneracies. We denote $X(d^i)$ by d_i, and call it a *face map*. Similarly, we denote $X(s^i)$ by s_i, and call it a *degeneracy map*. The term "simplicial operators" refers to faces, degeneracies, and in general any map in the image of a simplical set X.

Remark 3.1.3. Note that since $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$ is contravariant, face maps and degeneracy maps go in the opposite direction compared to the coface maps and codegeneracy maps in $\mathbf{\Delta}$. More precisely, $d_i : X_n \to X_{n-1}$ and $s_i : X_n \to X_{n+1}$.

Intuition 3.1.4. Note that a simplicial set is really just a function which takes each number $[n]$ and assigns it to a set X_n. Of course these sets have to satisfy some properties encoded by the faces and degeneracies, which is discussed in remark 3.1.7. The correct way to think about a simplicial set is thus as a collection of sets X_n for each n, which comes imbued with some extra structure given by the simplicial operators.

Remark 3.1.5. Since simplicial sets are just functors $\mathbf{\Delta}^{\mathrm{op}} \to \mathcal{S}\mathrm{et}$, we can construct a *category of simplicial sets* by simply considering the functor category $\mathrm{Fun}(\mathbf{\Delta}^{\mathrm{op}}, \mathcal{S}\mathrm{et}) = \mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}$. We will generally denote this category as $\mathrm{s}\mathcal{S}\mathrm{et}$. Note that morphisms in this category are given by natural transformations, so we define morphisms $\alpha : X_\bullet \to Y_\bullet$ of simplicial sets to simply be natural transformations from X to Y. By proposition 2.3.4, we only need to consider faces and degeneracies. We thus have that a natural transformation $\alpha : X_\bullet \to Y_\bullet$ is a collection of arrows $\alpha_n : X_n \to Y_n$ for each $[n] \in \mathbf{\Delta}$ such that the naturality squares

$$\begin{tikzcd} X_n \arrow[r, "\alpha_n"] \arrow[d, "d_i"'] & Y_n \arrow[d, "d_i"] \\ X_{n-1} \arrow[r, "\alpha_{n-1}"'] & Y_{n-1} \end{tikzcd} \qquad \begin{tikzcd} X_n \arrow[r, "\alpha_n"] \arrow[d, "s_i"'] & Y_n \arrow[d, "s_i"] \\ X_{n+1} \arrow[r, "\alpha_{n+1}"'] & Y_{n+1} \end{tikzcd}$$

commute. In other words, a morphism $\alpha : X_\bullet \to Y_\bullet$ is a collection of maps α_n for each n that commute with faces and degeneracies.

Terminology 3.1.6. Given a simplicial set $X_\bullet$, we call an element $x \in X_n$ an *n-simplex*. As in terminology 2.1.2, the plural of n-simplex is "n-simplices". We justify this terminology in remark 3.2.6. An n-simplex $x \in X_n$ is *degenerate* if it is in the image of *any* degeneracy map $s_i : X_{n+1} \to X_n$. Given an n-simplex $x \in X_n$, the ith *face* of x is the image of the face map $d_i(x)$.

Remark 3.1.7 (Simplicial identities)**.** We pause to investigate faces and degeneracies in simplicial sets. We have seen in proposition 2.3.4 and proposition 2.3.6, as summarized in remark 2.3.8, that the cofaces and codegeneracies, when subject to the cosimplicial operators, generate all morphisms in $\mathbf{\Delta}$. Similarly, any specified collection of faces $d_i : X_n \to X_{n-1}$ and degeneracies $s_i : X_n \to X_{n+1}$ will generate all simplicial operators for a simplicial set X, when subject to the following *simplicial identities*:

$$\begin{cases} d_i d_j = d_{j-1} d_i, & i < j \\ d_i s_j = s_{j-1} d_i, & i < j \\ d_j s_j = d_{j+1} s_j = 1 & \\ d_i s_j = s_j d_{i-1}, & i > j+1 \\ s_i s_j = s_{j+1} s_i, & i \leq j \end{cases}.$$

Note that these are simply the duals of proposition 2.3.6, and indeed they follow by contravariance and functoriality of X. Thus, to define a simplicial set X, it suffices to do the following three things:

1. Define a set X_n for each $[n] \in \mathbf{\Delta}$;
2. For each $i \in [n]$, define a map $d_i : X_n \to X_{n-1}$ and a map $s_i : X_n \to X_{n+1}$;
3. Show the maps of (2) satisfy the simplicial identities above.

These are precisely the qualities you obtain by forcing X to be functorial.

Exercise 3.1.8. We write the representable functor $\mathbf{\Delta}(-, [n])$ as Δ^n; the same notation for the standard n-simplex. We then write the Yoneda embedding $\mathbf{\Delta} \to \mathrm{s}\mathcal{S}\mathrm{et}$ as $\Delta^\bullet$. The reasoning for this notation will become clear in proposition 3.2.7, but for now, complete the following exercises.

1. Show that the n-simplicies $\Delta^0_n = *$ for all n, where $*$ denotes the singleton. In particular, show that Δ^0 is terminal in $\mathrm{s}\mathcal{S}\mathrm{et}$.
2. We call the simplicial set Δ^n the standard n-simplicial set, or sometimes the standard n-simplex, although we think the latter is stupid because it conflates notation even more with topological n-simplices. Yoneda's lemma tells us quite a lot about the standard simplicial sets for free. What can you deduce about these objects using only the Yoneda lemma, and basic category theory? Make sure part of your answer includes a description of faces and degeneracies for Δ^n.

Example 3.1.9. Any set X can be regarded as a simplicial set by taking each $X_n = X$, and by taking the faces and degeneracies to all be the identity map 1_X. This definition manifeslty satisfies the simplicial identities as stated in remark 3.1.7, indeed making X a simplicial set. We call such a simplicial set *discrete*, and this is the canonical way to view any set as a simplicial set. We will sometimes use sets in places where simplicial sets are supposed to be used. In those cases, we are implicitly considering the set X as a discrete simplicial set.

Definition 3.1.10. Let $X_\bullet$ be a simplicial set, and recall terminology 3.1.6.

1. The simplicial set $X_\bullet$ is said to be *n-skeletal* if for $m > n$, all m-simplices of $X_\bullet$ are degenerate.
2. The n-truncated simplex category $\mathbf{\Delta}_{\leq n}$ is the full subcategory of $\mathbf{\Delta}$ with objects $[m]$ for $m \leq n$.
3. The n-truncation $X_{\leq n}$ of a simplicial set $X_\bullet$ is the restriction of the simplicial set to the n-truncated simplex category $X|_{\mathbf{\Delta}_{\leq n}}$, which is intuitively understood as throwing away all dimensions above n.
4. The n-skeleton $\mathrm{sk}_n(X)$ of a simplicial set $X_\bullet$ is obtained by removing all nondegenerate m-simplices of X for $m > n$. This is often phrased as taking the n-truncation and replacing higher dimensions $m > n$ with freely generated degenerate simplices.

♡

3.2 Geometric realization

We were able to view objects in $\mathbf{\Delta}$ as geometric objects using the functor $\Delta : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{T}\mathrm{op}$. We can in fact achieve even better results with the *geometric realization functor* $|-| : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathcal{T}\mathrm{op}$. What follows in definition 3.2.1 is an extremely succinct way to summarize the geometric realization of a simplicial set using the language of *coends*.

Although it is simple, it is not particularly descriptive. Thus the reader who is unfamiliar with coends should not be concerned, as we will provide a much better description in construction 3.2.2 and remark 3.2.3. We found it suprisingly difficult to find a good resource on geometric realization; most sources expect us to already know it, yet we cannot find where we are supposed to know it from. We thank Emily Riehl for her presentation in [Rie11], and we thank Greg Friedman for his presentation in [Fri23].

Definition 3.2.1. Let $X_\bullet$ be a simplicial set, and let Δ^n denote the standard n-simplex as defined in definition 2.1.1. The geometric realization $|X_\bullet|$ of $X_\bullet$ is the coend

$$\int^{[n]\in\mathbf{\Delta}} X_n \times \Delta^n,$$

where X_n is given the discrete topology. This is equivalent to the definition via functor tensor products, since the usual tensor is the levelwise copower, and there is a homeomorphism $X_n \cdot \Delta^n \cong X_n \times \Delta^n$ when X_n is given the discrete topology. ♡

Construction 3.2.2. We now detail the construction of the geometric realization of a simplicial set. The experienced reader should be able to recognize this as the coend given in definition 3.2.1, computed via coproducts and coequalizers. Let $X_\bullet$ be a simplicial set, and recall that the coproduct in $\mathcal{T}\mathrm{op}$ is the disjoint union. The geometric realization $|X_\bullet|$ is computed by following these steps:

1. Construct the coproduct
$$\coprod_{[n]\in\mathbf{\Delta}} X_n \times \Delta^n,$$
where the set X_n is given the *discrete topology*, meaning all subsets $U \subseteq X_n$ are open.

2. We then impose an equivalence relation on the space, which we plan to quotient by. Let $\alpha : [m] \to [n]$ be a morphism in the simplex category $\mathbf{\Delta}$. Denote by α^* the morphism $X(\alpha)$, and denote by α_* the morphism $\Delta^\bullet(\alpha)$. We describe in more detail how to think of these morphisms in remark 3.2.3. We quotient by the following relation: for any $x \in X_n$ and any $p \in \Delta^m$,

$$(\alpha^*(x), p) \sim (x, \alpha_*(p)).$$

Note that this quotient only affects the parts of the coproduct indexed by $[n]$ and $[m]$, so we intuitively think of it as only quotienting those parts. We then impose this quotient *for all* morphisms α, which gives the geometric realization $|X_\bullet|$.

Remark 3.2.3. Of course by proposition 2.3.4, we only need to consider cofaces and codegeneracies, not *all* morphisms α in $\mathbf{\Delta}$. This simplifies the construction substantially, and we mainly described construction 3.2.2 in terms of all morphisms for pedagogical purposes; in particular to emphasize that cofaces and codegeneracies subsume all other morphisms in $\mathbf{\Delta}$. We continue with the understanding that we need only work with cofaces and codegeneracies.

- (faces) Recall that the action of X on the coface map $d^i : [n-1] \to [n]$ is given by the face map $d_i : X_n \to X_{n-1}$. Recall by definition 2.4.1 and intuition 2.4.4 that the corresponding coface map $d^i : \Delta^{n-1} \to \Delta^n$ inserts Δ^{n-1} at the ith face of Δ^n. What we want to do is think of each n-simplex $x \in X_n$ as identifying a geometric n-simplex via $\{x\} \times \Delta^n \subseteq X_n \times \Delta^n$. The product space can thus be thought of as an X_n-indexed (disjoint) collection of geometric n-simplices. The quotient process is thus interpreted as follows:
  1. Choose an n-simplex $x \in X_n$, and find the corresponding geometric n-simplex Δ^n.
  2. Find the ith face $d_i(x) \in X_{n-1}$ of x, and find the corresponding geometric $(n-1)$-simplex Δ^{n-1}.
  3. Glue the $(n-1)$-simplex Δ^{n-1} corresponding to $d_i(x)$ to the ith face of the geometric n-simplex Δ^n corresponding to x by gluing along the coface map $d^i : \Delta^{n-1} \to \Delta^n$.
- (degeneracies) Recall that the action of X on the codegeneracy $s^i : [n+1] \to [n]$ is given by the degeneracy map $s_i : X_n \to X_{n+1}$. Recall by definition 2.4.1 and intuition 2.4.4 that the corresponding codegeneracy map $s^i : \Delta^{n+1} \to \Delta^n$ identifies the ith and $(i+1)$th vertices of Δ^{n+1}, in effect "squishing" the simplex down to a degenerate n-simplex. Using the same logic as we did for faces, we note that each n-simplex $x \in X_n$ identifies a geometric n-simplex $\{x\} \times \Delta^n$, so the quotienting process is interpreted as follows:
  1. Choose an n-simplex $x \in X_n$, and find the corresponding geometric n-simplex Δ^n.
  2. Find a degenerate $(n+1)$-simplex $s_i(x) \in X_{n+1}$, and find the corresponding geometric $(n+1)$-simplex Δ^{n+1}.
  3. Collapse the $(n+1)$-simplex Δ^{n+1} corresponding to $s_i(x)$ to the n-simplex Δ^n corresponding to x along the ith codegeneracy s^i, identifying Δ^{n+1} with a degenerate simplex.

  We can summarize this process as taking all the geometric simplices corresponding to degenerate simplices, and turning them into degenerate simplices by composing along the corresponding codegeneracy.

The resulting information we gather from this analysis is twofold. Degenerate simplices can be thought of as *redundant information*, since the corresponding geometric simplex collapses down to a lower dimensional simplex. Faces of a simplex can be thought of as geometric faces of the corresponding geometric simplex, again by the process described above.

Intuition 3.2.4. The geometric realization, at first glance, is a fairly contrived and arbitrary construction. However, now that we've explored the construction in construction 3.2.2 and remark 3.2.3, we can provide some decent intuition for what simplicial sets really are.

Recall that in section 2.4, we obtained an understanding of how $\mathbf{\Delta}$ modeled simplicial sets and their relationships between each other in a combinatorial way by identifying each element with an n-simplex, and identifying each coface and codegeneracy with the corresponding geometric coface and codegeneracy. The category sSet is more powerful because it encodes *more* combinatorial information that $\mathbf{\Delta}$. In proposition 3.2.7, we will show the geometric realization of the standard n-simplicial sets are the standard n-simplices, and we will see even later that there is a notion of coface and codegeneracy between the standard n-simplicial sets, which correspond precisely to the notions of the same name between the standard geometric n-simplices. In this way, the objects Δ^n and the cofaces and codegeneracies between them encode the same information as $\mathbf{\Delta}$, since geometric realization gives us the same objects and the same morphisms.

However, s$\mathcal{S}$et is a much larger category than just the simplicial sets given by the Yoneda embedding. Other simplicial sets can have simpler structure, or more complicated structure, allowing us to build much more interesting spaces. Each simplicial set $X_\bullet$ encodes combinatorial information via its faces and degeneracies, and this information tells the geometric realization how to construct a geometric object. It starts by building geometric n-simplices for each n-simplex of $X_\bullet$, and then it glues them together based on the combinatorial information $X_\bullet$ contains. The vast amount of freedom we have when defining a simplicial set allows us to construct a large variety of different spaces. For example, example 3.3.5 constructs a space known as the "classifying space" for the very simple group $\mathbb{Z}/2\mathbb{Z} = \mathbb{Z}_2$, which is the geometric realization of a related simplicial set called the *nerve*. We find that the classifying space is the infinite real projective space $\mathbb{R}P^\infty$. Even though the group $\mathbb{Z}/2\mathbb{Z}$ is a very simple group, we are still able to construct an extremely complex space using the information it encodes. And we can also construct simple spaces with simplicial sets: in exercise 3.3.9 you will construct the discrete topological space on any set X in two different ways.

Thus, simplicial sets are ways to encode extremely complex data in a simple combinatorial fasion. It's a set of rules for gluing and collapsing geometric simplices, as described in construction 3.2.2, which can produce complex and important spaces. As we saw in section 2.4, this can simply quite a number of proofs, but simplicial objects have applications far beyond just this. In this paper, we will explore only a small number of the applications of simplicial sets, mainly with a focus on homotopy theory and category theory. However, applications of simplicial sets have become somewhat ubiquitous in algebraic topology, category theory, and fields beyond these in the past few years due to their *relatively* simple nature, compared to the complex structure they encode.

The reader at this point is likely hoping for some examples of geometric realization computations. However, the geometric realization can be fairly lengthy to compute for nontrivial cases, and many computations require theory that will not be developed until later, if at all. For this reason, most of our examples of computations will center around developing some of this theory, and will be sprinkled throughout the text. Our first computation will be proposition 3.2.7, where we compute the geometric realization of the standard n-simplicial sets Δ^n. For the reader who immediately wants to see an example of a more advanced worked geometric realization, we recommend skipping to section 3.3, and reading example 3.3.5.

Exercise 3.2.5. We claim in remark 3.2.3 that by proposition 2.3.4, we need only consider the faces and degeneracies when computing the geometric realization of a simplicial set. Make this notion precise by proving that if we construct the relations of construction 3.2.2 (2) for only the faces and degeneracies, then we already necessarily have the relation $(\alpha^*(x), p) \sim (x, \alpha_*(p))$ for any arrow α in $\mathbf{\Delta}$, with notation as in construction 3.2.2.

Remark 3.2.6. Throughout remark 3.2.3 and exercise 3.2.5, we continuously thought of each product $X_n \times \Delta^n$ not quite as a product space, but instead as an X_n-indexed collection of geometric n-simplices. The reader may be wondering if this is equivalent to the copower $X_n \cdot \Delta^n$, especially since these constructions are isomorphic as sets. Indeed, when X_n is given the discrete topology, the copower is homeomorphic to the product. This is not difficult to prove, and we encourage the reader to work out the proof if they are not convinced. We typically prefer product notation since it is more simple to write $(x, -)$ to denote that x is the indexing n-simplex in X_n, which simplifies the presentation of the relation we quotient by in construction 3.2.2. However this is largely notational, and we still think of it more as a copower.

Proposition 3.2.7. *The geometric realization of the standard n-simplicial set Δ^n, as defined in example 3.1.9, is the standard n-simplex Δ^n.*

Proof. Due to our overloaded notation, we will need to adopt some new notational conventions for the duration of this proof in order to make it readable. We will denote the standard n-simplicial as Δ^n, just like before, and will denote the standard geometric n-simplex as Δ_n, with a subscript. Any time both a superscript *and* a subscript show up of the form Δ^n_k, this denotes the k-simplices of Δ^n. Furthermore, the functor $\Delta_\bullet$ will denote the canonical embedding $\mathbf{\Delta} \to \mathcal{T}\mathrm{op}$, and $\Delta^\bullet$ will be the Yoneda embedding $\mathbf{\Delta} \to \mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}$.

An element of $|\Delta^n|$ is an equivalence class of a pair (f, p), where $f \in \Delta^n_k$ and $p \in \Delta^n$. By the defintion of

Δ^n as given in exercise 3.1.8, k-simplices are morphisms $f : [k] \to [n]$. Note that by remark 2.4.2, f induces a continuous map $\Delta_\bullet(f) : \Delta_k \to \Delta_n$. Since $p \in \Delta^k$, we can evaluate this morphism at p. Thus we propose the map $\varphi : |\Delta^n| \to \Delta_n$ given by $[(f,p)] \mapsto \Delta_\bullet(f)(p)$. It is not obvious that this map is well-defined, so before we can show it is a homeomorphism, we must show that any choice of representative of the equivalence class evaluates to the same point.

We first show the function is well-defined. Let $f \in \Delta^n_k$, $p \in \Delta_h$, and $\alpha : [h] \to [k]$ in $\mathbf{\Delta}$. Define the map ψ by

$$\begin{aligned} \coprod_{[k]\in\mathbf{\Delta}} \Delta^n_k \times \Delta_k &\xrightarrow{\ \psi\ } \Delta_n \\ (f,p) &\longmapsto \Delta_\bullet(f)(p). \end{aligned}$$

By construction 3.2.2, the equivalence relation we quotient by identifies points $(\alpha^*(f), p) \sim (f, \alpha_*(p))$, where α^* is the simplicial operator $\Delta^n_k \to \Delta^n_h$ induced by Δ^n, and $\alpha_* : \Delta_h \to \Delta_k$ is given by $\Delta_\bullet(\alpha)$. Recall that since Δ^n is given by the Yoneda embedding, the morphism α^* is simply given by precomposition, so we obtain that the relation reads as

$$(f\alpha, p) \sim (f, \Delta_\bullet(\alpha)(p)).$$

Applying ψ to both sides of this quotient gives, by functoriality of $\Delta_\bullet$,

$$\psi(f\alpha, p) = \Delta_\bullet(f\alpha)(p), \qquad \psi(f, (\Delta_\bullet(\alpha))(p)) = \Delta_\bullet(f)(\Delta_\bullet(\alpha)(p)) = (\Delta_\bullet(f) \circ \Delta_\bullet(\alpha))(p) = \Delta_\bullet(f\alpha)(p).$$

Since these are equal, we have that ψ identifies elements under the equivalence relation $\sim$, and thus by univeraslity of the quotient, there exists a unique dashed arrow

$$\begin{tikzcd} \coprod_{[k]\in\mathbf{\Delta}} \Delta^n_k \times \Delta_k \arrow[r, "\psi"] \arrow[d] & \Delta_n \\ |\Delta^n| \cong \left(\coprod_{[k]\in\mathbf{\Delta}} \Delta^n_k \times \Delta_k \right) /\sim \arrow[ur, dashed] & \end{tikzcd}$$

rendering the diagram commutative (in $\mathcal{S}\mathrm{et}$). By our extensive knowledge of universals in $\mathcal{S}\mathrm{et}$, we obtain an explicit description of the dashed arrow as $[(f,p)] = \psi(f,p) = \Delta_\bullet(f)(p)$, which is precisely the definition of φ. Thus φ arises by universality, and is therefore well-defined.

Now we show φ is continuous. Since φ arises by universality, and since by remark 3.0.1 all universals in $\mathcal{T}\mathrm{op}$ admit the same point-set definition as in $\mathcal{S}\mathrm{et}$, it suffices to show ψ is continuous, since the canonical quotient is continuous by definition. This will promote the universal problem to $\mathcal{T}\mathrm{op}$, rendering φ continuous, as required. We start by considering the restriction of ψ to only one component of the product $\Delta^n_k \times \Delta_k$. We can decompose the restriction of ψ as the composite

$$\Delta^n_k \times \Delta_k \xrightarrow{\Delta_\bullet \times 1} \mathcal{T}\mathrm{op}(\Delta_k, \Delta_n) \times \Delta_k \xrightarrow{\mathrm{eval}} \Delta_n.$$

If we give $\mathcal{T}\mathrm{op}(\Delta_k, \Delta_n)$ the compact-open topology, then it is well known that the evaluation map $\mathrm{eval}(\Delta_\bullet(f), p) = \Delta_\bullet(f)(p)$ is continuous whenever the domain (in this case, Δ_k) is locally compact Hausdorff. Indeed, all Euclidean spaces $\mathbb{R}^n$ are locally compact Hausdorff by Heine-Borel, and all open or closed subsets of locally compact Hausdorff spaces are locally compact Hausdorff. Since n-simplices are clearly closed as subspaces of $\mathbb{R}^{n+1}$, this follows. It suffices to show $\Delta_\bullet \times 1$ is continuous. Note that $\Delta_\bullet \times 1$ arises as the unique dashed arrow in the diagram

$$\begin{tikzcd} \Delta^n_k \arrow[r, "\Delta_\bullet"] & \mathcal{T}\mathrm{op}(\Delta_k, \Delta_n) \\ \Delta^n_k \times \Delta_k \arrow[u] \arrow[d] \arrow[r, dashed, "\Delta_\bullet \times 1"] & \mathcal{T}\mathrm{op}(\Delta_k, \Delta_n) \times \Delta_k \arrow[u] \arrow[d] \\ \Delta_k \arrow[r, "1"'] & \Delta_k \end{tikzcd}$$

induced by universality of products in $\mathcal{T}$op. Since the identity 1 is continuous, it suffices to show $\Delta_\bullet$ is continuous. Recall that all functions whose domain is a discrete topological space are continuous, since the preimage of any open set is a set, and any set is open in a discrete space. Recall in definition 3.2.1 and construction 3.2.2 that we grant the set of k-simplices Δ^n_k the discrete topology, and thus $\Delta_\bullet$ is continuous. We thus have that the restriction of ψ is continuous for any k.

Since every restriction $\psi|_k$ of ψ along the k-indexed component of the coproduct is continuous, universality of coproduct assembles the arrows $\psi|_k$ into a unique dashed arrow as indicated

$$\begin{tikzcd} \Delta^n_k \times \Delta_k \arrow[d] \arrow[dr, "\psi|_k"] & \\ \coprod_{[k]\in\mathbf{\Delta}} \Delta^n_k \times \Delta_k \arrow[r, dashed] & \Delta_n \end{tikzcd}$$

rendering the diagram commutative. This unique dashed arrow is precisely ψ since precomposing with the injection $\Delta^n_k \times \Delta_k$ is the same as restricting along the k-indexed component. Thus, ψ arises by universality in $\mathcal{T}$op, and is therefore continuous. We thus obtain that φ is continuous, as required.

We now show φ is an isomorphism by constructing an explicit inverse. Define the map η as the composite

$$\begin{aligned} \Delta_n &\longrightarrow \coprod_{[k]\in\mathbf{\Delta}} \Delta^n_k \times \Delta_k \longrightarrow |\Delta^n| \\ p &\longmapsto (1, p) \in \Delta^n_n \times \Delta_n \longmapsto [(1, p)]. \end{aligned}$$

To be fully clear, we are mapping the point p to the point $p \in \Delta_n$ in the geometric n-simplex indexed by the identity map $1 \in \Delta_n$. First, we show this map is continuous. Note that given the diagram of solid arrows

$$\begin{tikzcd} & \Delta^n_n & & \\ \Delta_n \arrow[ur, "x\mapsto 1"] \arrow[r, dashed] \arrow[dr, "1"'] & \Delta^n_n \times \Delta_n \arrow[u] \arrow[d] \arrow[r, "i_n"] & \coprod_{[k]\in\mathbf{\Delta}} \Delta^n_k \times \Delta_k \arrow[r, "\pi"] & |\Delta^n| \\ & \Delta_n & & \end{tikzcd}$$

universality of product induces a unique dashed arrow as indicated, rendering the diagram commutative. We also clearly see that η can be defined as the composite along the main row, where π denotes the canonical projection and i_n denotes the inclusion into the coproduct at the n-indexed component. Since π and i_n are continuous, we simply need to show the dashed arrow is continuous, and by universality it suffices to show the identity and the constant map at 1 are both continuous. Both are clearly continuous, giving us that η is continuous, as required.

Now we show η is an inverse of φ. It's clearly a right inverse of φ:

$$(\varphi\eta)(p) = \varphi[(1, p)] = \Delta_\bullet(1)(p) = 1(p) = p,$$

because $\Delta_\bullet$ is a functor and thus preserves identities. To show it is a left inverse, note that

$$(\eta\varphi)[(f, p)] = \eta(\Delta_\bullet(f)(p)) = [(1, \Delta_\bullet(f)(p))].$$

We thus need to show that $(f, p) \sim (1, \Delta_\bullet(f)(p))$. Indeed, since $f = f^*(1)$, we have

$$(f, p) = (f^*(1), p) \sim (1, \Delta_\bullet(f)(p)),$$

concluding the proof. ■

Remark 3.2.8. Proposition 3.2.7 justifies our overloaded notation for Δ^n and $\Delta^\bullet$. Because of this fact, many authors prefer to denote the standard geometric n-simplex by $|\Delta^n|$. We prefer this notation as well, but both choices overload notation in some way (until we prove proposition 3.2.7 at least), and we felt that for a basic introduction, our choice of notation was, for lack of a better word, less bad. However, now that we have proven proposition 3.2.7, **we now prefer to switch to the more common notation** of denoting the standard geometric n-simplex by $|\Delta^n|$.

Remark 3.2.9. Geometric realizations of simplicial sets are always compactly generated, and in particular are always CW complexes with one n-cell for each nondegenerate n-simplex of the simplicial set. This is extremely convenient, since the category $\mathcal{CG}$ is notably much better behaved than the general category $\mathcal{T}op$, while still containing most notable topological spaces, making working in the category of compactly generated spaces particularly appealing.

3.3 The nerve of a category

Given a category $\mathcal{C}$, there is an extremely important construction known as the *nerve* of $\mathcal{C}$, which encodes *all* of the information of $\mathcal{C}$ into a simplicial set. This is an extremely important construction for a number of reasons. For example, it's proven in [Lur09, Prop. 1.1.2.2][4] that all nerves are ∞-categories, and in particular the nerve is the canonical way of regarding categories as ∞-categories. Although exploring this precisely is beyond the scope of this paper, we trust that the reader understands this is an important result, even if they do not quite understand it. With this in mind, we present the definition of the nerve.

Definition 3.3.1. The nerve $N(\mathcal{C})$ of a category $\mathcal{C}$ is the simplicial set defined as follows.

1. n-simplices are composable n-tupes of morphisms in $\mathcal{C}$. More precisely, the set $N(\mathcal{C})_n$ of n-simplices is the set of all n-tuples $(f_0, \ldots, f_{n-1})$ of composable arrows between $(n+1)$ objects

$$c_0 \xrightarrow{f_0} c_1 \xrightarrow{f_1} \cdots \xrightarrow{f_{n-2}} c_{n-1} \xrightarrow{f_{n-1}} c_n.$$

   The set of 0-simplices is thus simply the collection of objects $\mathrm{Obj}(\mathcal{C})$.

2. Degeneracy maps $s_i : N(\mathcal{C})_n \to N(\mathcal{C})_{n+1}$ are defined by inserting the identity morphism as the ith arrow. Viewed as a n-tuple, this looks like the map

$$(f_0, \ldots, f_{i-1}, f_i, f_{i+1}, \ldots, f_{n-1}) \mapsto (f_0, \ldots, f_{i-1}, 1_{c_i}, f_i, f_{i+1}, \ldots, f_{n-1}).$$

   Viewed as a diagram, this looks like taking the diagram of composable arrows

$$c_0 \longrightarrow \cdots \xrightarrow{f_{i-1}} c_i \xrightarrow{f_i} \cdots \longrightarrow c_n$$

   and mapping it to the diagram

$$c_0 \longrightarrow \cdots \xrightarrow{f_{i-1}} c_i \xrightarrow{1_{c_i}} c_i \xrightarrow{f_i} \cdots \longrightarrow c_n.$$

3. Face maps have different defintions for "inner" and "outer" faces. An inner face is a map d_i where $0 < i < n$, and an outer face is d_i where $i = 0$ or $i = n$.
   (a) Inner faces $d_i : N(\mathcal{C})_n \to N(\mathcal{C})_{n-1}$ are defined by composing the ith and the $(i+1)$th morphisms. Viewed as a map of n-tupes, this is the map

$$(f_0, \ldots, f_{i-1}, f_i, f_{i+1}, \ldots, f_{n-1}) \mapsto (f_0, \ldots, f_{i-1}, f_{i+1} \circ f_i, \ldots, f_{n-1}).$$

   Viewed as a diagram, this looks like taking the diagram of composable arrows

$$c_0 \longrightarrow \cdots \longrightarrow c_{i-1} \xrightarrow{f_i} c_i \xrightarrow{f_{i+1}} c_{i+1} \longrightarrow \cdots \longrightarrow c_n$$

   and mapping it to the diagram

$$c_0 \longrightarrow \cdots \longrightarrow c_{i-1} \xrightarrow{f_{i+1} \circ f_i} c_{i+1} \longrightarrow \cdots \longrightarrow c_n.$$

[4] Joyal proved this first, but I prefer Lurie's treatment. However, this fact is so fundamental that it's likely a proof may date back to Boardman and Vogt, but I unfortunatey don't have access to their paper.

(b) Outer faces are defined by deleting a morphism at the end of the chain. The 0th face deletes the first morphism, and the nth face deletes the last morphism. For example, if

$$c_0 \longrightarrow c_1 \longrightarrow \cdots \longrightarrow c_{n-1} \longrightarrow c_n$$

is a diagram of composable arrows in $\mathcal{C}$, then the 0th face maps it to

$$c_1 \longrightarrow \cdots \longrightarrow c_{n-1} \longrightarrow c_n$$

and the nth face maps it to

$$c_0 \longrightarrow c_1 \longrightarrow \cdots \longrightarrow c_{n-1}.$$

♡

Example 3.3.2. Consider the poset category $\mathbf{n}$. A 0-simplex k in the nerve $N(\mathbf{n})$ is simply an object of $\mathbf{n}$, equivalently a non-negative integer $k < n$. A 1-simplex f in the nerve is a single morphism in $\mathbf{n}$, which corresponds to a relation $h \leq k$ for $h, k \in \mathbf{n}$. Thus, 1-simplices are relations $h \leq k$ for objects h, k. 2-simplices are pairs of morphisms $h \xrightarrow{f} k \xrightarrow{g} \ell$, which correspond to relations $h \leq k \leq \ell$. In general, we find that m-simplices are relations

$$h_0 \leq h_1 \leq \cdots \leq h_m,$$

where $h_i \in \mathbf{n}$ for all i.

Since $\mathbf{n}$ is well-ordered, we can say slightly more about it. Recall that terminology 3.1.6 defined *degenerate* simplices as simplices which are the image of a degeneracy. In view of definition 3.3.1, degeneracies in the nerve simply add identity arrows. Thus any m-simplex with an identity arrow is automatically a degenerate simplex, since by removing the identity arrow we get an $(m-1)$-simplex, and the image of that $(m-1)$-simplex under the proper degeneracy map simply re-inserts the identity morphism back where it was. Since there are no non-identity endomorphisms in $\mathbf{n}$, note that the longest possible chain of relations in $\mathbf{n}$ without using an identity arrow is the sequence of relations

$$0 \leq 1 \leq 2 \leq \cdots \leq n-2 \leq n-1.$$

This sequence of morphisms has length $n-1$, and it is the unique chain of arrows of length $n-1$ that can be formed without using identities. Thus, $N(\mathbf{n})$ has precisely 1 nondegenerate $(n-1)$-simplex. Moreover, no arrows longer than length $n-1$ can be made without using identities, so $N(\mathbf{n})$ has *no* nondegenerate k-simplices for $k \geq n$.

Example 3.3.3. Let G be a group. You have likely seen that we can view a group G as a category $\mathbf{B}G$ by taking $\mathbf{B}G$ to have only one object, usually denoted $*$, by taking each morphism of $\mathbf{B}G$ to be an element of G, and by taking composition in $\mathbf{B}G$ to be the group operation in G. In particular, every morphism in $\mathbf{B}G$ has an inverse by the group axioms for G. We call categories where all morphisms are invertible *groupoids*, as an analogue of a group. Most authors prefer to identify a group G with its category, and will refer to both of them simply as G. Some authors prefer to write $\mathbf{B}G$ for the category, since the category can be identified with it's *delooping* in ∞-$\mathcal{G}$rpd. Although we prefer the former notation, we believe the latter reduces the opportunity for confusion, so we use it here for pedagogical purposes.

We can compute the nerve of a group G by considering it as a category. Since $\mathbf{B}G$ has one object, $N(\mathbf{B}G)_0 = \{*\}$. The 1-simplices are the morphisms, which are precisely the elements of G. Thus, $N(\mathbf{B}G)_1 = G$. The 2-simplices are chains of two morphisms, which are equivalently just pairs of elements in G, so we obtain $N(\mathbf{B}G)_2 = G \times G$. This very clearly continues to give us $N(\mathbf{B}G)_n = G^n$, where G^n is the n-repeated product. Indeed, this even generalizes to the 0-simplices, since the empty product in $\mathcal{S}$et is the terminal object, which is the singleton. Similarly to example 3.3.2, we see that degeneracies insert the identity element $e \in G$, so degenerate simplices in $N(\mathbf{B}G)$ are simply tuples that contain the identity element.

Exercise 3.3.4. Compute the nerves of a small groupoid $\mathcal{G}$ and the simplex category $\mathbf{\Delta}$. Where possible, give descriptions of the degenerate simplices.

Example 3.3.5. The geometric realizations of the nerves of various categories are oftentimes fairly important objects. For example, the geometric realization of the nerve of a group is the *classifying space* BG of a group G. We will compute the very simple example $B\mathbb{Z}_2$ by following example 3.3.2. We begin by constructing the coproduct

$$\coprod_{[n]\in\mathbf{\Delta}} \mathbb{Z}_2^n \times |\Delta^n|.$$

We will start by working only with lower dimension terms in the coproduct, to get a sense of how it will work. Since each $x \in \mathbb{Z}_2^n$ indexes a geometric n-simplex, we will adopt the convention of writing $|\Delta^n_{(-)}|$ to denote that the geometric n-simplex $|\Delta^n|$ is indexed by the n-simplex $(-)$. Since $|\mathbb{Z}_2| = 2$, we have that the order $|\mathbb{Z}_2^n| = 2^n$. We thus have

$$\coprod_{[n]\in\mathbf{\Delta}} \mathbb{Z}_2^n \times |\Delta^n| \cong \left(|\Delta^0_*|\right) \amalg \left(|\Delta^1_0| \amalg |\Delta^1_1|\right) \amalg \left(|\Delta^2_{(0,0)}| \amalg |\Delta^2_{(0,1)}| \amalg |\Delta^2_{(1,0)}| \amalg |\Delta^2_{(1,1)}|\right) \amalg \cdots$$

We begin by quotienting by degeneracies. Since $0 \in \mathbb{Z}_2$ is the identity, any geometric simplex indexed by a tuple containing 0 is indexed by a degenerate simplex. For example, the simplex $|\Delta^2_{(0,0)}|$ is index by $(0,0) = s_0(0) = s_1(0)$, so $|\Delta^2_{(0,0)}|$ is identified with $|\Delta^1_0|$. This means that we should identify these geometric simplices.

I claim we can simply drop the higher dimension simplex in the coproduct up to homeomorphism. The proof is somewhat involved. By universality of coproduct, we can simply consider the space $|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|$, since a given homeomorphism, along with the identity arrows for all other maps, assemble into a homeomorphism by universality. We want to quotient this space by the relation $(s_0(x), p) \sim (x, s^0(p))$ for $x \in \mathbb{Z}_2^1$ and $p \in |\Delta^2_{(0,0)}|$. Note that the following proof is invariant under whether or not we choose the 0th or 1st degeneracy, so I chose the 0th. Our choice of x is 0, since x is simply the indexing simplex for $|\Delta^1_0|$, and each point p can be represented as a triple (t_0, t_1, t_2) of nonnegative numbers which sum to 1. Since $s_0(0) = (0,0)$, and since $s^0(t_0, t_1, t_2) = (t_0 + t_1, t_2)$, we thus have the relation

$$(0, (t_0 + t_1, t_2)) \sim ((0,0), (t_0, t_1, t_2)).$$

We want to show there is a homeomorphism

$$\left(|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|\right)/\sim \;\cong |\Delta^1_0|.$$

To prove this, define the map

$$\begin{aligned} f : \left(|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|\right)/\sim &\longrightarrow |\Delta^1_0| \\ [(0, (t_0, t_1))] &\longmapsto (0, t_0, t_1) \end{aligned}$$

We must first show this map is well-defined. In particular, we must show each class $[(0, (t_0, t_1))]$ has precisely one representative of the given form. To do this, note that the class admits the definition as

$$[(0, (t_0, t_1))] = \{(0, t_0, t_1)\} \cup \left\{(p_0, p_1, p_2) \in \Delta^2_{(0,0)} \mid p_0 + p_1 = t_0, \text{ and } p_2 = t_1\right\}.$$

Since we are required to fix $p_0 + p_1 = t_0$ and $p_2 = t_1$, clearly this equivalence class only has one representative of the given form, so the map is indeed well-defined. To show the map is continuous, we note that we can view the map as a map

$$|\Delta^1_0| \amalg |\Delta^2_{(0,0)}| \longrightarrow |\Delta^1_0|$$

by noting that

- the map is the identity on the $|\Delta^1_0|$ component of the coproduct,
- and the map is s^0 on,

so universality of coproduct assembles our map via the unique dashed arrow in the diagram of solid arrows

$$\begin{tikzcd}
{|\Delta^1_0|} \arrow[d] \arrow[dr, "1"] & \\
{|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|} \arrow[r, dashed] & {|\Delta^1_0|} \\
{|\Delta^2_{(0,0)}|} \arrow[u] \arrow[ur, "s^0"'] &
\end{tikzcd}$$

rendering the diagram commutative. This map manifestly identifies elements which are related to each other under the relation $\sim$, so it factors through the quotient

$$\begin{tikzcd}
{|\Delta^1_0|} \arrow[d] \arrow[dr] \arrow[drr, bend left, "1"] & & \\
{|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|} \arrow[r, dashed] & {\left(|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|\right)/\sim} \arrow[r, dashed, "f"] & {|\Delta^1_0|} \\
{|\Delta^2_{(0,0)}|} \arrow[u] \arrow[ur] \arrow[urr, bend right, "s^0"'] & &
\end{tikzcd}$$

Since we see that our map f arises from universality conditions between continuous maps, and since $\mathcal{T}\mathrm{op}$ is complete and cocomplete, necessarily f is continuous.

It thus suffices to show f has a continuous inverse. The inverse is obvious, we simply define $g : (0,(t_0,t_1)) \mapsto [0,(t_0,t_1)]$, and this is automatically well-defined since we showed earlier each equivalence class has precisely one representative of this form. f and g are manifeslty inverse functions, so we now show g is continuous. Note that g arises equivalently as the composite

$$|\Delta^1_0| \xrightarrow{\ i\ } |\Delta^1_0| \amalg |\Delta_{(0,0)}| \xrightarrow{\ \pi\ } \left(|\Delta^1_0| \amalg |\Delta^2_{(0,0)}|\right) ,$$

where i is the inclusion and π is the natural projection onto the quotient. Since g is a composite of continuous maps, it is indeed continuous.

We have thus shown that we may freely drop degenerate simplices (almost... see remark 3.3.6) in the geometric realization, since that is precisely the same as quotienting by degeneracies. Thus quotienting by degeneracies is homeomorphic to the space

$$\coprod_{[n]\in\mathbf{\Delta}} |\Delta^n|.$$

We now need to quotient by faces. We can show that every single non-degenerate simplex is a face of a single non-degenerate simplex in precisely two ways. Let $(1,\ldots,1)_{n-1}$ be the nondegenerate $(n-1)$-simplex, and let $(1,\ldots,1)_n$ be the nondegenerate n-simplex. Taking an inner face map on $(1,\ldots,1)_n$ corresponds to composing two morphisms, or in other words, taking the group operation on two elements. Since $1+1=0$, this can be viewed as removing two 1s and replacing it with a 0, turning it into a degenerate simplex. However, the outer faces simply remove one of the outer 1s, and thus map $(1,\ldots,1)_n \mapsto (1,\ldots,1)_{n-1}$. Thus the only faces that are not accounted for by the degeneracy quotient are the outer faces.

Since we only have a single copy of each geometric n-simplex left, from now on we will denote elements $p \in |\Delta^n|$ as (n,p); in other words, the first slot in the pair is the dimensionality of the simplex the second slot lives in as a point.

We begin by working inductively, and we start with the 0-skeleton $|\Delta^0|$, which is the singleton $*$. Moving on to the 1-skeleton, we see that the inner face maps

$$|\Delta^0| \underset{d^1}{\overset{d^0}{\rightrightarrows}} |\Delta^1|$$

insert $*$ at 0 and 1 in the 1-simplex. Quotienting thus identifies $(1,0) \sim * \sim (1,1)$; in other words we have identified the endpoints of the 1-simplex. Since the 1-simplex is simply a line segment, this construction is homeomorphic to the unit circle S^1. Thus, the 1-skeleton is simply S^1. Now we move on to the 2-skeleton. The inner faces (subject to the previous quotient) $d^0, d^2 : S^1 \to |\Delta^2|$ insert the 1-skeleton at the 0th and 2nd faces. Gluing these faces together along the circle is precisely the antipodal map, producing the real projective plane $\mathbb{R}P^2$. One can induct on this construction and obtain that the n-skeleton is $\mathbb{R}P^n$. The geometric realization can then be seen as the colimit over the inclusions of the skeletons

$$|N(\mathbf{B}G)| = \operatorname{colim}(X_0 \hookrightarrow X_1 \hookrightarrow X_2 \hookrightarrow \cdots),$$

which is precisely the definition of $\mathbb{R}P^\infty$. Thus, $B\mathbb{Z}_2 = \mathbb{R}P^\infty$.

Remark 3.3.6. The reader might think that the section of example 3.3.5 stating that we may drop degenerate simplices likely generalizes, since we rarely appealed to the specific problem of example 3.3.5 in that section. Although our proof is sufficient for the purposes of example 3.3.5, it fails in general due to transfinite considerations. We proved in example 3.3.5 that for any degenerate n-simplex, we may drop the corresponding geometric n-simplex. Note that this only proves that we may drop one at a time. We can induct on the argument and find that we may drop any finite number k of degenerate simplices at a time, but this still fails in general since a simplicial set need not have finitely many degenerate n-simplices. The proof only held for example 3.3.5 since there are $|\mathbb{Z}_2^n| - 1 = 2^n - 1$ degenerate n-simplices, which is clearly less than ∞, and inducting on n gives us that we may drop these for each X_n.

The general result is stated in proposition 3.3.7, but the proof is notably longer. For this reason, we delegate the proof to appendix A for the interested reader.

Proposition 3.3.7. *Let $X_\bullet$ be a simplicial set. If X_n^{nd} is the set of nondegenerate n-simplices of $X_\bullet$, define the equivalence relation $\sim_f$ on $\coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n|$ as the relation generated by the rule $(y, \mu_*(p)) \sim_f (z, \beta_*(p))$ if*

- *μ is a composite of face maps,*
- *β is a composite of codegeneracy maps,*
- *z is nondegenerate,*
- *$\beta^*(z) = \mu^*(y)$.*

Let $\sim$ be the equivalence relation of construction 3.2.2 Then there is a homeomorphism

$$|X_\bullet| = \left(\coprod_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n| \right) /\!\sim\; \cong \left(\coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n| \right) / \sim_f .$$

Proof. Proven in appendix A. ■

Corollary 3.3.8. *Let $X_\bullet$ be an k-skeletal simplicial set, as defined in definition 3.1.10. Then its geometric realization $|X_\bullet|$ is homeomorphic to the geometric realization of its k-truncation $|X_{\leq k}|$:*

$$|X_{\leq k}| := \int^{[n]\in\mathbf{\Delta}_{\leq k}} X_n \times |\Delta^n| \cong \left(\coprod_{n\leq k} X_n \times |\Delta^n| \right) / \sim .$$

For those unfamiliar with coends, the only change here is that the limit is taken over $\mathbf{\Delta}_{\leq k}$ as defined in definition 3.1.10.

Exercise 3.3.9. Fun fact: we never proved the nerve was a simplicial set. Prove that the faces and degeneracies of the nerve as defined in definition 3.3.1 satisfy the simplicial identities (remark 3.1.7). Afterwards, let X be a set, and recall by example 3.1.9 that we may view X as a simplicial set. We may also view it as a category by considering the discrete category structure on X. Compute the nerve $N(X)$, then apply corollary 3.3.8 to show $|N(X)| \cong |X| \cong X$, where the latter X is viewed as a topological space with the discrete topology.

Exercise 3.3.10 (Advanced)**.** Prove that the geometric realization of the nerve of the poset category $\mathbf{n} = [n+1]$ is the geometric realization of the standard $(n+1)$-simplicial set $|N(\mathbf{n})| \cong \left|\Delta^{n+1}\right|$.

3.4 Boundaries and horns

We now introduce the notion of a simplicial subset, along with two important constructions known as the *horn*, and the *boundary/simplicial sphere*, which are of particular importance in a variety of contexts.

Definition 3.4.1. Let $X_\bullet$ be a simplicial set. A *simplicial subset* of $X_\bullet$ is a simplicial set $Y_\bullet$ where

- For all n, the n-simplices of $Y_\bullet$ are a subset of those of $X_\bullet$, meaning $Y_n \subseteq X_n$;
- The faces and degeneracies in $Y_\bullet$ agree with those in $X_\bullet$; in particular, the faces d_i^Y of $Y_\bullet$ are defined as the restrictions of the faces d_i^X and degeneracies s_i^X of $X_\bullet$ along the subset $Y_n \subseteq X_n$

$$d_i^Y = d_i^X|_{Y_n}, \qquad s_i^Y = s_i^X|_{Y_n}.$$

We write $Y_\bullet \subseteq X_\bullet$ to denote that it is a simplicial subset. ♡

Remark 3.4.2. Unfortunately, not all collections of subsets $Y_n \subset X_n$ of a simplicial set $X_\bullet$ form a simplicial subset. Yes, one can define the faces and degeneracies for $Y_\bullet$ by restricting those of $X_\bullet$ along the relevant subsets, but *the image of each restriction must be contained in the corresponding subset Y_n*.

For example, consider the discrete simplicial set $X_\bullet$ where $X_n = \mathbb{Z}$ for all n, and all faces and degeneracies are the identities. We can construct subsets $Y_\bullet$ by taking $Y_n = X_n$ for all n *except* for $n = 1$, where we take $Y_n = \mathbb{N}$. It doesn't matter whether or not we choose $0 \in \mathbb{N}$ here. By definition 3.4.1, for $Y_\bullet$ to be a simplicial subset, the faces and degeneracies must agree with those of $X_\bullet$; in particular they are the restrictions along the subsets $Y_\bullet$. Since $Y_0 = X_0$, the face $d_0 : Y_0 \to Y_1$ must be the same. Since the face is the identity, it maps each element to itself. Note that $-1 \in Y_0 = \mathbb{Z}$, and $d_0(-1) = -1 \notin Y_1 = \mathbb{N}$. Since the image is not in the subset Y_1, $Y_\bullet$ cannot be a simplicial subset, since the definition of $Y_\bullet$ as given does not form a valid simplicial set.

Remark 3.4.3. In contrast to remark 3.4.2, it turns out that all one has to check to determine whether an object $Y_\bullet$ is a simplicial subset of a simplicial set $X_\bullet$ is

- $Y_n \subseteq X_n$ for all n;
- The faces and degeneracies agree with X;
- The faces and degeneracies are well-defined in the sense of remark 3.4.2, meaning the sets Y_n contain the images of all faces and degeneracies.

This is fairly easily shown. The first two criteria guarentee that if $Y_\bullet$ is a simplicial set, it will be a simplicial subset of $X_\bullet$, so we need only check that it is a simplicial set. Since the faces and degeneracies are all well-defined, remark 3.1.7 tells us that we need only check the faces and degeneracies satisfy the simplicial identities. This is manifestly true: the faces and degeneracies in $Y_\bullet$ are simply restrictions of those on $X_\bullet$, and since $X_\bullet$ is a simplicial set, the faces and degeneracies satsify the simplicial identities on the whole of $X_\bullet$. This means they must satisfy them for every element of each X_n, and thus every subset of each X_n.

In particular, we see that to specify a simplicial subset $Y_\bullet$ of $X_\bullet$, one needs only specify the sets Y_n, and then show these sets contain the images of the restricted faces and degeneracies, since the definitions of the faces and degeneracies are forced by definition 3.4.1.

Definition 3.4.4. Let $\alpha : X_\bullet \to Y_\bullet$ be a morphism of simplicial sets. The *image* of α, denoted $\operatorname{im} \alpha_\bullet$, is defined as the simplicial subset of $Y_\bullet$ given by

$$(\operatorname{im} \alpha)_n = \operatorname{im}(\alpha_n : X_n \to Y_n),$$

where α_n is the component of α at n. We make use of the convenient notation and write $\operatorname{im}\alpha_n$ to denote the set of n-simplices of $\operatorname{im}\alpha$. ♡

Claim 3.4.5. *With notation as in definition 3.4.4, the image is indeed a simplicial subset of $Y_\bullet$.*

Proof. By remark 3.4.3, we need only show the images of each face and degeneracy land within the specified sets. We start by showing this for degeneracies. Let $y \in \operatorname{im}\alpha_n$, and we want to show $s_i^{\operatorname{im}\alpha}(y) \in \operatorname{im}\alpha_{n+1}$. Recall by remark 3.1.5 that morphisms of simplicial sets are natural tranformations, meaning they commute with degeneracies as indicated in the diagram

$$\begin{tikzcd} X_n \arrow[r, "\alpha_n"] \arrow[d, "s_i^X"'] & Y_n \arrow[d, "s_i^Y"] \\ X_{n+1} \arrow[r, "\alpha_{n+1}"'] & Y_{n+1} \end{tikzcd}$$

By definition, for each $y \in \operatorname{im}\alpha_n$, there exists at least one $x \in X_n$ such that $y = \alpha_n(x)$. Since the degeneracy in $\operatorname{im}\alpha_\bullet$ is the same as the degeneracy for $Y_\bullet$, we have

$$s_i^{\operatorname{im}\alpha}(y) = s_i^Y(y) = \left(s_i^Y\alpha_n\right)(x) = \left(\alpha_{n+1}s_i^X\right)(x),$$

where the final step follows by commutativity of the diagram. By definition, this is in the image of α_{n+1}, completing the proof for degeneracies. The proof for faces is the same, and is done by constructing the corresponding naturality diagram for faces instead of degeneracies. ■

Definition 3.4.6. Let $X_\bullet, Y_\bullet$ be simplicial sets, and suppose that for each n, the faces and degeneracies of $X_\bullet$ and $Y_\bullet$ agree on the intersection $X_n \cap Y_n$. In symbols,

$$s_i^Y|_{X_n\cap Y_n} = s_i^X|_{X_n\cap Y_n}, \qquad d_i^Y|_{X_n\cap Y_n} = d_i^X|_{X_n\cap Y_n}.$$

Then the *union* $X_\bullet \cup Y_\bullet$ of $X_\bullet$ and $Y_\bullet$ is defined as the levelwise union

$$(X \cup Y)_n = X_n \cup Y_n,$$

and the faces and degeneracies are defined by

$$s_i(x) = \begin{cases} s_i^X(x), & x \in X \\ s_i^Y(x), & x \in Y \end{cases}, \qquad d_i(x) = \begin{cases} d_i^X(x), & x \in X \\ d_i^Y(x), & x \in Y \end{cases}.$$

♡

Claim 3.4.7. *With notation as in definition 3.4.6, $X_\bullet \cup Y_\bullet$ is a simplicial set. In particular, $X_\bullet, Y_\bullet$ are simplicial subsets of their union.*

Proof. The fact that $X_\bullet, Y_\bullet \subseteq X_\bullet \cup Y_\bullet$ follows manifestly from the definition, provided we indeed show $X_\bullet \cup Y_\bullet$ is a simplicial set, which is not terribly difficult. As usual, we appeal to remark 3.1.7 and show that the faces and degeneracies are well defined, and satisfy the simplicial identities. Since $X_\bullet$ and $Y_\bullet$ agree on their intersection, for any $x \in X_n \cap Y_n$, we must have $d_i^X(x) = d_i^Y(x)$, and the same for degeneracies. Thus, the maps are well-defined. To show they satisfy the simplicial identities, note that for all $x \in X_n \cup Y_n$, we have $x \in X_n$ or $x \in Y_n$. Without loss of generality, let $x \in X_n$. Since $X_\bullet$ is a simplicial set, the simplicial operators satisfy the simplicial identities for x. Extending this argument to $Y_\bullet$, we find that the simplicial operators satisfy the simplicial identities for all $x \in X_n \cup Y_n$, and thus $X_\bullet \cup Y_\bullet$ is indeed a simplicial set. ■

Previously in terminology 2.1.6, we defined the *face* of the standard n-simplex, and we were even able to encode this information into the category $\mathbf{\Delta}$ in intuition 2.4.4. As explained in exercise 3.2.5, the combinatorial and geometric information encoded by simplicial sets is in many ways more general than that of $\mathbf{\Delta}$, and as seen in proposition 3.2.7, the standard n-simplicial sets encode the information of the standard geometric n-simplex. We should thus think that there should be some way to define the faces of the standard n-simplicial set. Indeed there is such a way to do so, and the definition of the face of the standard n-simplicial set will give rise to much more important constructions.

Definition 3.4.8. Let d^i be the image of the coface map $[n-1] \to [n]$ under the Yoneda embedding. More precisely, let $d^i : \Delta^{n-1} \to \Delta^n$ denote precomposition by the coface map $d^i : [n-1] \to [n]$.

1. The ith face $\partial_i \Delta^n$ of the simplical set Δ^n is the simplicial subset $\operatorname{im} d^i_\bullet$.
2. The *boundary* $\partial \Delta^n$ of the standard n-simplicial set Δ^n, also called the *simplicial n-sphere*, is the simplicial subset
$$\partial \Delta^n \coloneqq \bigcup_{i \in [n]} \partial_i \Delta^n = \bigcup_{i \in [n]} \operatorname{im} d^i_\bullet \subseteq \Delta^n.$$
3. The *kth horn* Λ^k_n of the standard n-simplicial set is the simplicial subset
$$\Lambda^k_n \coloneqq \bigcup_{i \in [n], i \neq k} \partial_i \Delta^n = \bigcup_{i \in [n], i \neq k} \operatorname{im} d^i_\bullet \subseteq \partial \Delta^n \subseteq \Delta^n.$$

♡

Remark 3.4.9. Because boundaries and horns are defined as unions over simplicial subsets of Δ^n, their faces and degeneracies automatically agree with Δ^n, and in particular agree with each other, making the union of simplicial sets well-defined.

Intuition 3.4.10. The geometric intuition for faces has already been discussed. Our definition of faces implies that the coface maps $d^i : \Delta^{n-1} \to \Delta^n$ do the "simplicial set equivalent" of inserting the standard $(n-1)$-simplicial set at the "ith face of the standard n-simplicial set". It remains to be shown of course that the geometric realization of $\operatorname{im} d^i_\bullet$ is the ith face of $|\Delta^n|$, but this will be shown soon.

For now, working with the understanding that $\operatorname{im} d^i_\bullet$ represents the face of the standard n-simplicial set, we define the *boundary* of the standard n-simplicial set as the union of all of its faces. We expect a union of simplicial sets to correspond to a union of geometric objects, so working under this intuition, we would imagine the geometric realization of the boundary $\partial \Delta^n$ to be the physical boundary of the standard n-simplex. Once again, we will show this to be true shortly.

The only definition that may not be immediately clear is that of a horn. The definition of the kth horn Λ^k_n is the union of all the faces of Δ^n *except* the kth one. For example, consider Λ^2_2. Geometrically, this corresponds to taking the standard 2-simplex, deleting the interior so that we are only left with its faces (which constitute the boundary), and then deleting the 2nd face. Figure 6 illustrates this geometrically. The remaining two faces, f_0 and f_1, assemble into a sort of "$\wedge$" shape, which looks somewhat like a two-dimensional horn. Hence the namesake. It remains to be shown of course that the geometric realizations align with these objects. We can in fact prove stronger results. In section 4.1, we will show that geometric realization is a left adjoint functor, and thus preserves colimits. It thus suffices to show that unions and images can be viewed as colimits, and our intuition here will follow as corollary 3.4.13.

Proposition 3.4.11. *Let $F : \mathcal{J} \to \mathcal{D}^{\mathcal{C}}$ be a diagram with $\mathcal{D}$ complete, and let $\operatorname{ev}_c : \mathcal{D}^{\mathcal{C}} \to \mathcal{D}$ be the evaluation functor $F \mapsto F(c)$, $\alpha \mapsto \alpha_c$. Then $\lim F$, if it exists, is given by $(\lim F)(c) = \lim(\operatorname{ev}_c \circ F)$.*

Proof. See [ML98, §5.3 Thm. 1]. ■

Corollary 3.4.12. *Images and unions of simplicial sets, as defined in definition 3.4.1 and definition 3.4.6, are colimits in* sSet*. In particular, if $X_\bullet, Y_\bullet \in \mathrm{s}\mathcal{S}\mathrm{et}$ and $\alpha : X_\bullet \to Y_\bullet$ is a morphism of simplicial sets, then*

1. $|\operatorname{im}(\alpha : X_\bullet \to Y_\bullet)_\bullet| \cong \operatorname{im}(\alpha : |X_\bullet| \to |Y_\bullet|)$*, and*
2. $|X_\bullet \cup Y_\bullet| \cong |X_\bullet| \cup |Y_\bullet|$.

Proof. Recall that $\mathrm{s}\mathcal{S}\mathrm{et} = \mathcal{S}\mathrm{et}^{\mathbf{\Delta}^{\mathrm{op}}}$ is a functor category, and since $\mathcal{S}\mathrm{et}$ is complete and cocomplete, proposition 3.4.11 and its dual apply. In particular, any diagram F in sSet must satisfy $(\lim F)_n = \lim(\operatorname{ev}_n \circ F)$. This tells us that limit in sSet are computed levelwise, meaning in particular

1. Let $\alpha : X_\bullet \to Y_\bullet$ be a morphism of simplicial sets. The *limit* definition of the image $\operatorname{im} \alpha_\bullet$ (not definition 3.4.4) must satisfy $(\operatorname{im} \alpha)_n = \operatorname{im}(\alpha_n)$, where the second is a limit in $\mathcal{S}\mathrm{et}$. Note that this is precisely definition 3.4.4, and thus we have that our image in sSet can be regarded as a limit. Since $\mathcal{S}\mathrm{et}$ admits canonical decompositions for morphisms, there is an isomorphism $\operatorname{im}(\alpha_n) \cong \operatorname{coim}(\alpha_n)$. The coimage is a colimit, and thus we may regard the image in sSet as isomorphic to a colimit.

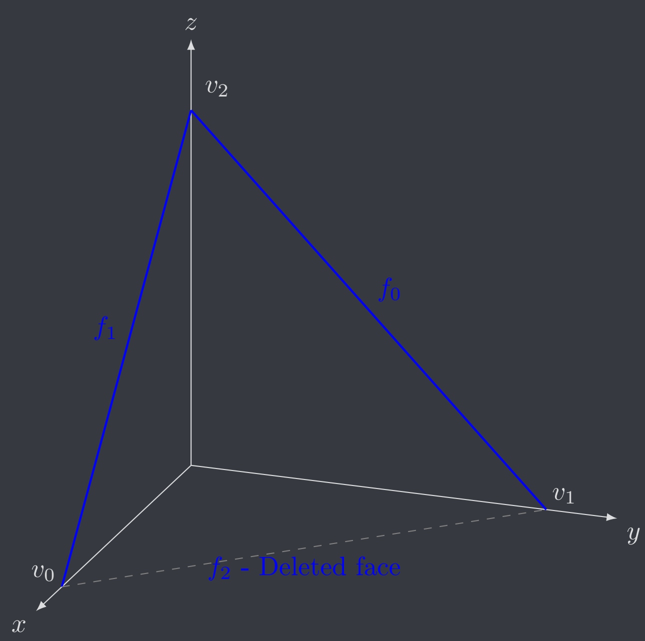

Figure 6: The second horn $|\Lambda_2^2|$ of $|\Delta^2|$

2. Let $X_\bullet, Y_\bullet$ be simplicial sets whose simplicial operators agree on their intersection. Because their simplicial operators agree, it is easy to see that $X_\bullet \cap Y_\bullet$ is a subsimplicial set of both $X_\bullet$ and $Y_\bullet$, with the intersection given by levelwise intersection. Note that the pushout $Z_\bullet$ of $X_\bullet$ and $Y_\bullet$ along the canonical inclusions $X_\bullet \cap Y_\bullet \hookrightarrow X_\bullet, Y_\bullet$ can be computed levelwise, as indicated in the diagram

$$\begin{tikzcd} (X \cap Y)_n \arrow[r, hook] \arrow[d, hook] \arrow[dr, phantom, "\lrcorner"] & X_n \arrow[d] \\ Y_n \arrow[r] & Z_n \end{tikzcd}$$

in $\mathcal{S}\mathrm{et}$. It is well known that this pushout can be given by $Z_n = X_n \cup Y_n$, which is precisely the definition of the n-simplices of the union $(X \cup Y)_n$. Thus, we may regard the union as the pushout, which is in particular a colimit.

Recall that geometric realiation $|-|$ is left adjoint, and thus preserves colimits. This implies that $|X_\bullet \cup Y_\bullet| \cong |X_\bullet| \cup |Y_\bullet|$ and that $|\operatorname{coim}\alpha| = \operatorname{coim}|\alpha|$. Note that $\mathcal{T}\mathrm{op}$ also admits canonical decompositions of morphisms, so we obtain $\operatorname{coim}|\alpha| \cong \operatorname{im}|\alpha|$, giving a string of isomorphism

$$|\operatorname{im}\alpha| \cong |\operatorname{coim}\alpha| \cong \operatorname{coim}|\alpha| \cong \operatorname{im}|\alpha|.$$

■

Corollary 3.4.13. *The following statements hold.*

1. *The geometric realization of the ith face is the ith face of the geometric n-simplex:*

$$|\partial_i \Delta^n| \cong d^i(|\Delta^{n-1}|).$$

2. *The geometric realization of the boundary is the boundary of the geometric n-simplex:*

$$|\partial \Delta^n| \cong \partial\, |\Delta^n|\,.$$

3. *The geometric realization of the kth horn is the kth horn (defined as the union of all faces other than k) of the geometric n-simplex:*

$$\left|\Lambda_n^k\right| \cong \bigcup_{i \in [n], i \neq k} \operatorname{im}(d^i : \left|\Delta^{n-1}\right| \to |\Delta^n|).$$

Proof. Indeed, these are all special cases of *corollary* 3.4.12, and it suffices to show $|d^i| = d^i : |\Delta^{n-1}| \to |\Delta^n|$. This is shown in **??**. ■

Notation 3.4.14. As is likely expected, we represent the geometric kth horn of the standard n-simplex as $|\Lambda_n^k|$. We also adopt the convention of denoting the ith face of the geometric n-simplex as $\partial_i|\Delta^n|$.

Remark 3.4.15. It should be fairly clear to anyone who has taken topology that the boundary of the triangle $\partial|\Delta^2|$ is homeomorphic to the 1-sphere S^1. Similarly, it's clear that the boundary of the tetrahedron $\partial|\Delta^3|$ is homeomorphic to the 2-sphere S^2. Indeed this generalizes: for $n > 1$, there is a homeomorphism $\partial|\Delta^n| \cong S^{n-1}$. For this reason, many authors call $\partial\Delta^n$ the simplicial n-sphere.

4 Homotopy

4.1 The singular complex functor

Definition 4.1.1. Let X be a topological space. The *singular set* of X is the simplicial set $\mathrm{Sing}_\bullet(X)$ given by

- $\mathrm{Sing}_n(X) = \mathcal{T}\mathrm{op}(|\Delta^n|, X)$;
- Degeneracies and faces are given by precomposition:

$$d_i : \mathcal{T}\mathrm{op}(|\Delta^n|, X) \xrightarrow{(d^i)^*} \mathcal{T}\mathrm{op}(\left|\Delta^{n-1}\right|, X)$$
$$s_i : \mathcal{T}\mathrm{op}(|\Delta^n|, X) \xrightarrow[(s^i)^*]{} \mathcal{T}\mathrm{op}(\left|\Delta^{n+1}\right|, X).$$

♡

Remark 4.1.2. It is immediately checked that this is a simplicial set since $\mathrm{Hom}(-, x)$ is contravariant for all x in all categories $\mathcal{C}$, and since the geometric cofaces and codegeneracies satsify the cosimplicial identities, precomposition must satisfy the faces and degeneracies. One may also view this as a consequence of Yoneda's lemma.

Proposition 4.1.3. *The singular complex is functorial* $\mathrm{Sing} : \mathcal{T}\mathrm{op} \to \mathrm{s}\mathcal{S}\mathrm{et}$. *Action on morphisms is given by levelwise postcomposition.*

Proof. The singular complex is functorial levelwise, meaning for each n, there is a functor $\mathrm{Sing}_n : \mathcal{T}\mathrm{op} \to \mathcal{S}\mathrm{et}$ given by $\mathrm{Sing}_n = \mathcal{T}\mathrm{op}(|\Delta^n|, -)$. One might hope that these assemble into a functor $\mathcal{T}\mathrm{op} \to \mathrm{s}\mathcal{S}\mathrm{et}$. Note that for any $f : X \to Y$ in $\mathcal{T}\mathrm{op}$, the map induced at each level is simply postcomposition $\mathcal{T}\mathrm{op}(|\Delta^n|, f) = f_*$. It's well-known that $(fg)_* = f_* g_*$ and that 1_* is the identity by functoriality of $\mathrm{Hom}(x, -)$, so we need only show f_* is a morphism of simplicial sets. By remark 3.1.5, we must show the diagrams

$$\begin{tikzcd}
\mathcal{T}\mathrm{op}(|\Delta^x|, X) \arrow[r, "f_*"] \arrow[d, "(d^i)^*"'] & \mathcal{T}\mathrm{op}(|\Delta^n|, Y) \arrow[d, "(s^i)^*"] \\
\mathcal{T}\mathrm{op}(\left|\Delta^{n-1}\right|, X) \arrow[r, "f_*"'] & \mathcal{T}\mathrm{op}(\left|\Delta^{n-1}\right|, Y)
\end{tikzcd}
\qquad
\begin{tikzcd}
\mathcal{T}\mathrm{op}(|\Delta^n|, X) \arrow[r, "f_*"] \arrow[d, "(s^i)^*"'] & \mathcal{T}\mathrm{op}(|\Delta^n|, Y) \arrow[d, "(d^i)^*"] \\
\mathcal{T}\mathrm{op}(\left|\Delta^{n+1}\right|, X) \arrow[r, "f_*"'] & \mathcal{T}\mathrm{op}(\left|\Delta^{n+1}\right|, Y)
\end{tikzcd}$$

commute. These diagrams boil down to the equalities

$$f_*(d^i)^* = (d^i)^* f_*, \qquad f_*(s^i)^* = (s^i)^* f_*,$$

which are easily verified by tracing a map g around the squares, and noting that we are precomposing by the same degeneracies and faces in both verticals of each square. ■

Notation 4.1.4. Many authors prefer to write $S(X)$ for the singular set, and $S_n(X)$ for its set of n-simplices. We prefer writing Sing, since we think it is less ambiguous and because we think it looks cooler.

The singular complex functor is of great importance for a variety of reasons. For example, singular sets are always *Kan complexes*, which have significant homotopic importance and are a model for ∞-groupoids; there is an adjunction $|-| \dashv \mathrm{Sing}$ which is a *Quillen equivalence*; and in particular the geometric realization of a singular set is homotopy equivalent to the original topological space. For now, we explore a more simple application of the singular complex functor which also has extreme importance in many fields: the singular homology of a topological space.

Remark 4.1.5. In construction 4.1.6, we will consider *simplicial abelian groups*, which is a contravariant functor $X : \mathbf{\Delta}^{\mathrm{op}} \to \mathcal{A}\mathrm{b}$. More explicitly, they are groups X_n for each $n \geq 0$, and *group homomorphism* $d_i : X_n \to X_{n-1}$ and $s_i : X_n \to X_{n+1}$ satisfying the simplicial identities (remark 3.1.7). This definition is precisely the same as a simplicial set, but using $\mathcal{A}\mathrm{b}$ instead of $\mathcal{S}\mathrm{et}$. One can equivalently define a simplicial object in a category $\mathcal{C}$ as a contravariant functor $\mathbf{\Delta}^{\mathrm{op}} \to \mathcal{C}$.

Construction 4.1.6 (Singular homology)**.** Let X be a topological space. Definition 4.1.1 gives the singular set $\mathrm{Sing}_\bullet(X) \in \mathrm{s}\mathcal{S}\mathrm{et}$. Recall that the free abelian group is a functor $\mathbb{Z}[-] : \mathcal{S}\mathrm{et} \to \mathcal{A}\mathrm{b}$ which maps sets to abelian groups generated by that set, and maps functions to the group homomorphism induced by that function on basis elements. Denote by $\mathrm{s}\mathcal{A}\mathrm{b}$ the category of simplicial abelian groups. We can extend the free abelian group functor to a functor $\mathbb{Z}[-] : \mathrm{s}\mathcal{S}\mathrm{et} \to \mathrm{s}\mathcal{A}\mathrm{b}$ by applying it levelwise:

$$\mathbb{Z}[\mathrm{Sing}(X)]_n = \mathbb{Z}[\mathrm{Sing}_n(X)], \qquad d_i^{\mathbb{Z}} = \mathbb{Z}[d_i^{\mathrm{Sing}}].$$

We will denote by $C_n(X)$ the abelian group $\mathbb{Z}[\mathrm{Sing}_n(X)]$.

Given the simplicial abelian group $C_\bullet(X)$, let $d_i : C_n \to C_{n-1}$ be the face map, and define the *boundary morphism* on the basis simplices $\sigma \in \mathrm{Sing}_n(X)$ as

$$\begin{aligned} \partial_n : C_n &\longrightarrow C_{n-1} \\ \sigma &\longmapsto \sum_{i=0}^{n} (-1)^i d_i(\sigma) \end{aligned}$$

and we extend it to all of $C_\bullet$ via linearity. We can offer a good description of what $d_i(\sigma)$ is defined as. On basis simplices, the face maps of the simplicial abelian group $C_\bullet(X)$ must agree with the face map $d_i : \mathrm{Sing}_n(X) \to \mathrm{Sing}_{n-1}(X)$ by the universal property of free abelian groups, and the face map d_i in singular sets is precomposition by the coface map d^i of geometric simplices. Thus, $d_i(\sigma) = \sigma \circ d^i \in \mathrm{Sing}_{n-1}(X)$ also holds between C_n and C_{n-1}, so we will also write

$$\partial_n : \sigma \mapsto \sum_{i=0}^{n} (-1)^i (\sigma \circ d^i)$$

on basis simplices.

We claim for now that $\partial_{n-1}\partial_n = 0$. We prove this in proposition 4.1.8, but the proof is lengthy, so we defer it for now. We thus have that $(C_\bullet(X), \partial_\bullet)$ is a chain complex of abelian groups. The assignment $C_\bullet(X) \mapsto (C_\bullet(X), \partial_\bullet)$ is functorial, which is explored in remark 4.1.9. We can thus compute the homology groups, typically denoted $H_\bullet(X)$ rather than $H_\bullet(C(X))$, which we recall are defined as

$$H_n(X) = \frac{\operatorname{Ker} \partial_{n-1}}{\operatorname{Im} \partial_n}.$$

These homology groups are the *singular homology* of the topological space X, and are the canonical and most useful construction of homology for topological spaces.

Remark 4.1.7. Construction 4.1.6 generalizes in a variety of ways. Firstly, we do not have to work with abelian groups. Given a (unital) ring R, we can define the R-valued homology R-modules $H_\bullet(X; R)$ by simply taking the free R-module on $\mathrm{Sing}_\bullet(X)$, rather than the free abelian group, and proceeding with proposition 4.1.3 in the same way. Of course, abelian groups are simply $\mathbb{Z}$-modules, so we have $H_\bullet(X; \mathbb{Z}) = H_\bullet(X)$. The other generalization is the notion of simplicial homology, which defines homology for a fairly broad class of simplicial sets known as *simplicial complexes*. We will not introduce this notion in this paper in favor of exploring homotopical implicications of the functor Sing, but the interested reader is encouraged to see **??**, or to take a class on algebraic topology.

Proposition 4.1.8. *With notation as in construction 4.1.6, $\partial_{n-1}\partial_n = 0$. In particular, $C_\bullet(X)$ is a chain complex.*

Proof. This proof is most straightforward when appealing to the simplicial identities of remark 3.1.7, rather than to the explicit description of simplicial operators in $C_\bullet(X)$. For this reason, let d_i denote the ith face in the simplicial abelian group $C_\bullet(X)$, and let $\sigma \in C_n(X)$ be an n-simplex. We have

$$\sigma \xmapsto{\partial_n} \sum_{i=0}^{n}(-1)^i d_i(\sigma) \xmapsto{\partial_{n-1}} \sum_{j=0}^{n-1}(-1)^i d_j\left(\sum_{i=0}^{n}(-1)^i d_i(\sigma)\right).$$

Since d_j is a group homomorphism, it distributes over addition, giving

$$\sum_{j=0}^{n-1}(-1)^j d_j\left(\sum_{i=0}^{n}(-1)^i d_i(\sigma)\right) = \sum_{j=0}^{n-1}(-1)^j \sum_{i=0}^{n} d_j\left((-1)^i d_i(\sigma)\right).$$

Note that $(-1)^i x$ is either x or $-x$. For the former, $d_j((-1)^i x) = d_j(x)$, and for the latter, $d_j((-1)^i x) = d_j(-x) = -d_j(x)$, since d_j is a group homomorphism. We may thus factor out the $(-1)^i$, giving

$$\sum_{j=0}^{n-1}(-1)^j \sum_{i=0}^{n} d_j\left((-1)^i d_i(\sigma)\right) = \sum_{j=0}^{n-1}(-1)^j \sum_{i=0}^{n}(-1)^i d_j d_i(\sigma) = \sum_{j=0}^{n-1}\sum_{i=0}^{n}(-1)^j(-1)^i d_j d_i(\sigma).$$

a ■

Remark 4.1.9. show functoriality

4.2 The adjunction $|-| \dashv \mathrm{Sing}$

4.3 Quillen's model structure

see [Hov99] for good source, we follow a lot. og is [Qui67] tho

4.4 Simplicial homotopy

at the end i should recommend goerss n jardine

5 Closing remarks

i wanna write some closing remarks on like where the reader can go next to continue learning if they so desire. for example altho i bashed it in the intro, [GJ99] is solid for the reader interested in simplicial homotopy, and the reason i bashed it is cuz like it offers basically 0 geometric intuition so its aids as a first resource for simplicial sets. for the reader interested in more general homotopy theory of course we recommend something in higher category theory right? but idk what to recommend i mean [Rie14] is okay but i really dont think its very good, [Lur09] is prolly out of the reach of the reader despite what Lurie says in [Lur11], but what else do i say? i would say the reader should understand some of the theory of model cats before HTT so maybe i recommend exerpts of a model cat paper, hovey §3 is good for Quillen's model structure and Joyals is introduced in HTT §2.2.5 using ∞-categorical machinery so maybe just recommend that? gotta think on this also i should have recommendations for the topologically-motivated reader who for some reason didnt read [Fri23] instead

A Proof of proposition 3.3.7

Lemma A.0.1. *Let $x \in X_\bullet$ be a degenerate simplex. Then there is a unique nondegenerate simplex $y \in X_k$ for $k \leq n$, and a unique composite of codegeneracies $\alpha : [n] \to [k]$ in $\mathbf{\Delta}$ such that $\alpha^*(x) = y$, where α^* is the simplicial operator induced by α in the simplicial set $X_\bullet$.*

Proof. First we show existence. Let $x \in X_n$ be degenerate. By definition, there exists $y \in X_{n-1}$ and $i \leq n$ such that $s_i(y) = x$. If y is nondegenerate, we are done. Otherwise, we continue inductively with the same argument. Note that there is an upper bound; in particular all 0-simplices are nondegenerate since no degeneracies have domain X_0. Since n is finite, induction gives us the required existence.

Uniqueness of the map is obtained by noting any composite of degeneracies is a composite of codegeneracies in $\mathbf{\Delta}$, which is a surjective map, and as a subresult of proposition 2.3.4, surjections are *uniquely* determined by their degenerate decomposition. Uniqueness of the nondegenerate simplex is obtained as follows: let $\alpha^*(y) = \alpha^*(z) = x$. Since α^* is a composite of degeneracies, α is a surjection in $\mathbf{\Delta}$, which means it admits a right inverse by the axiom of choice. Denote this right inverse by σ. We prove by contradiction that σ is order preserving. Note that $\sigma : [k] \to [n]$. If $k = 0$, then σ is trivially order preserving, so we may freely assume there exist distinct $a, b \in [k]$. In particular, we may let $a < b$ strictly, and suppose that $\sigma(a) \geq \sigma(b)$. Then since α is order preserving, $\alpha\sigma(a) \geq \alpha\sigma(b)$, but since $\alpha\sigma = 1$, we have that $a \geq b$, a contradiction. Thus, σ is order preserving, and a morphism in $\mathbf{\Delta}$. It induces a simplicial operator σ^* in $X_\bullet$, which must satisfy

$$z = (\alpha\sigma)^*(z) = \sigma^*\alpha^*(z) = \sigma^(x) = \sigma^*\alpha^*(y) = (\alpha\sigma)^*(y) = y.$$

Thus, the nondegenerate simplex is unique. ∎

Remark A.0.2. We would like to quotient only by faces, but the issue with quotienting only by $(d_i(x), p) \sim (x, d^i(p))$ is that faces of a nondegenerate simplex need not be nondegenerate ([Fri23]; see the discussion following definition 3.8), and thus wouldn't be present in the new space, so the relation wouldn't be very well-defined. We instead pass to the relation $\sim_f$, which is defined as the relation generated by the following rule.

Let X_n^{nd} denote the set of *nondegenerate* n-simplices of $X_\bullet$. Define the equivalence relation $\sim_f$ on $\coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n|$ by $(x, \mu_*(p)) \sim_f (y, \beta_*(p))$ if

- μ is a composite of coface maps;
- the unique nondegenerate simplex of $\mu^*(x)$ given by lemma A.0.1 is y;
- and β is the unique composite of codegeneracies given by lemma A.0.1 such that $\beta^*(y) = \mu^*(x)$.

Note that μ can be the identity by choosing it to be the empty composite, which is an empty composite of coface maps, giving

$$(x, p) \sim_f (x, p),$$

since x is already nondegenerate and thus the corresponding composite of codegeneracies is also empty, hence another identity. Since the relation is reflexive, we simply demand symmetry and transitivity, giving us the equivalence relation by generating it on the above rule.

Proposition A.0.3 (Prop. 3.3.7). *Let $X_\bullet$ be a simplicial set, let $\sim_f$ be as defined in remark A.0.2, let $\sim$ be the relation given in construction 3.2.2 (2), and let X_n^{nd} denote the set of nondegenerate n-simplices of X. Then there is a homeomorphism*

$$|X_\bullet| := \left(\coprod_{n\geq 0} X_n \times |\Delta^n|\right) /\!\sim\ \cong \left(\coprod_{n\geq 0} X_n^{nd} \times |\Delta^n|\right) / \sim_f .$$

In words, one may construct the geometric realization by deleting nondegenerate simplices, and quotienting by the relation of remark A.0.2, which is extremely similar to quotienting purely by faces.

Proof. We want to show

$$\left(\coprod_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n|\right) /\!\sim\ \cong \left(\coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n|\right) /\!\sim_f,$$

Appealing to lemma A.0.1, we define the map

$$\begin{aligned} \varphi : \coprod_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n| &\longrightarrow \coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n| \\ (x,p) = (\alpha^*(y), p) &\longmapsto (y, \alpha_*(p)), \end{aligned}$$

where α^* is as defined in lemma A.0.1, and $\alpha_* = |\Delta^\bullet|(\alpha) : |\Delta^k| \to |\Delta^n|$. We defer showing the map is continuous until the end of the proof. Our goal for now will be to show that if $x \sim y$ in the domain of φ, then $\varphi(x) \sim_f \varphi(y)$ in the codomain; postcomposing with the quotient morphism will then identify these elements, and thus the composite identifies all elements under $\sim$, hence we may invoke universality.

First, we show it respects degeneracies. We can show a strict equality here, thus the relation follows by reflexivity. Let $x \in X_n$ and $p \in |\Delta^{n+1}|$, such that $(s_i(x), p) \sim (x, s^i(p))$. By lemma A.0.1, let $y \in X_k$ be nondegenerate, and let $\alpha^*(y) = x$. By uniqueness of α, we have that $s_i\alpha^*(y)$ is the unique composition of degeneracies mapping $y \mapsto x$. Note also that $s_i\alpha^* = X(\alpha s^i)$ by contravariance of $X_\bullet$, and thus the induced morphism on geometric simplices is $|\Delta^\bullet|(\alpha s^i) = \alpha_* s^i$ by functoriality. We have

$$\varphi(s_i(x), p) = \varphi(s^i\alpha^*(y), p) = (y, \alpha_* s^i(p)), \qquad \varphi(x, s^i(p)) = \varphi(\alpha^*(y), s^i(p)) = (y, \alpha_* s^i(p)).$$

Thus, φ respects $\sim$ for degeneracies.

We now check faces. Recall that we must show that $\varphi(d_i(x), p) \sim_f \varphi(x, d^i(p))$. Keep notation as before, meaning $x = \alpha^*(y)$ is given by lemma A.0.1. We have

$$\varphi(x, d^i(p)) = \varphi(\alpha^*(y), d^i(p)) = (y, \alpha_* d^i(p)).$$

The other side is more difficult. Note that $(d_i(x), p) = (d_i\alpha^*(y), p) = ((\alpha d^i)^*(y), p)$ by contravariance of $X_\bullet$. By proposition 2.3.4, αd^i admits a unique factorization into a composite of codegeneracies (a epi component) and a composite of cofaces (a monic component). Write $\alpha d^i = \mu\eta$ for this monic-epi decomposition, where μ is the monic and η is the epi. We thus have

$$d_i(x) = d_i\alpha^*(y) = (\alpha d^i)^*(y) = (\mu\eta)^*(y) = \eta^*\mu^*(y).$$

Since $\mu^*(y)$ may be degenerate, we define $\mu^*(y) = \beta^*(z)$, where z is the unique nondegenerate simplex and β is the unique composite of degeneracies, as given in lemma A.0.1. By uniqueness, we have that $d_i(x) = \eta^*\beta^*(z) = (\beta\eta)^*(z)$ is the unique composite of degeneracies for $d_i(x)$ given by lemma A.0.1, so

$$\varphi(d_i(x), p) = \varphi((\beta\eta)^*(z), p) = (z, \beta_*\eta_*(p)).$$

Returning to $(x, d^i(p))$, we note that since $d^i\alpha = \mu\eta$, since $\beta^*(z) = \eta^*(y)$, and since μ is a composite of cofaces, we have

$$\varphi(x, d^i(p)) = (y, \alpha_* d^i(p)) = (y, \mu_*\eta_*(p)) \sim_f (z, \beta_*\eta_*(p)) = \varphi(d_i(x), p)$$

by definition of $\sim_f$ as given in remark A.0.2.

We have thus shown $(x,p) \sim (y,q)$ implies $\varphi(x,p) \sim_f \varphi(y,q)$. Since the quotient map π_{nd} equalizes $\sim_f$, have that $\pi_{nd}\varphi$ descends along the quotient

$$\begin{array}{ccc} \displaystyle\coprod_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n| & \xrightarrow{\quad\varphi\quad} & \displaystyle\coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n| \\ {\scriptstyle\pi}\big\downarrow & & \big\downarrow{\scriptstyle\pi_{nd}} \\ \left(\displaystyle\coprod_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n|\right)/\!\sim & \overset{}{\underset{\psi}{\dashrightarrow}} & \left(\displaystyle\coprod_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n|\right)/\!\sim_f \\ & & \\ [(\alpha_*(y), p)] & \underset{\psi}{\longmapsto} & [(y, \alpha^*(p))]_f \end{array}$$

as the unique dashed arrow, rendering the diagram commutative. Since each $X_n^{nd} \times |\Delta^n|$ admits a continuous inclusion functor into $X_n \times |\Delta^n|$, by universality these assemble into the hooked solid arrow in the diagram

$$\begin{tikzcd}
\coprod\limits_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n| \arrow[d, "\pi"'] & \coprod\limits_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n| \arrow[l, hook', "i"] \arrow[d, "\pi_{nd}"] \\
\left(\coprod\limits_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n|\right)/\!\sim & \left(\coprod\limits_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n|\right)/\!\sim_f \arrow[l, dashed, "\eta"']
\end{tikzcd}$$

Note that π identifies all elements related by $\sim$, and thus we would like to show that $(x,p) \sim_f (y,q)$ implies $i(x,p) \sim i(y,q)$. This is indeed true, but we defer the proof to lemma A.0.4. Since π identifies elements under $\sim$, the composite πi identifies elements under $\sim_f$, and by universality of quotient, there is a unique dashed arrow ζ as indicated, rendering the diagram commutative. We now show ψ and ζ are inverses. One direction is clear. Let (x,p) be a point in the (not quotiented) disjoint union of nondegenerate simplices. Then x is already nondegenerate, so $\alpha^* = 1$, giving us $\varphi i(x,p) = \varphi(x,p) = (x,p)$. For the converse, let $x = \alpha^*(y)$ be the representation of lemma A.0.1, and note that

$$i\varphi(x,p) = i(y, \alpha_*(p)) = (y, \alpha_*(p)) \sim (\alpha^*(y), p) = (x,p).$$

Thus, $i\varphi$ descends to $\zeta\psi = 1$ along the quotient. We thus have that these maps are inverses, so we must now show they are continuous.

We already know ζ is continuous since inclusions are continuous. To show ψ is continuous, recall that since X_n has the discrete topology, the product is isomorphic to the copower

$$X_n \times |\Delta^n| \cong X_n \cdot |\Delta^n| \cong \coprod_{x \in X_n} \{x\} \times \Delta^n.$$

Let $x = \alpha^*(y)$ as given in lemma A.0.1. Note that $x \mapsto y$ is continuous since X_n is discrete, and α_* is continuous by functoriality of $|\Delta^\bullet|$, so by universality of coproduct, given the diagram of solid arrows

$$\begin{tikzcd}
\{x\} \times |\Delta^n| \arrow[r, "(x\mapsto y)\times\alpha_*", "z"'] \arrow[d] & \{y\} \times |\Delta^n| \arrow[d] \\
\coprod\limits_{x\in X_n} \{x\} \times |\Delta^n| \arrow[r, dashed, "\coprod z"'] & \coprod\limits_{x\in X_n^{nd}} \{x\} \times |\Delta^n|
\end{tikzcd}$$

there exists a unique dashed arrow as indicated, rendering the diagram commutative. Note that we have chosen to denote $(x \mapsto y) \times \alpha_*$ by z for simplicity. Since the diagram is in $\mathcal{T}\mathrm{op}$, note that the dashed arrow is in particular continuous. Recalling our isomorphism above, once again universality of coproduct assembles the solid arrows

$$\begin{tikzcd}
X_n \times |\Delta^n| \arrow[r, "\coprod z"] \arrow[d] & X_n^{nd} \times |\Delta^n| \arrow[d] \\
\coprod\limits_{[n]\in\mathbf{\Delta}} X_n \times |\Delta^n| \arrow[r, "\coprod\coprod z"'] & \coprod\limits_{[n]\in\mathbf{\Delta}} X_n^{nd} \times |\Delta^n|
\end{tikzcd}$$

into a unique dashed arrow as indicated, rendering the diagram commutative. In particular, the dashed arrow $\coprod\coprod z$ is continuous. Inspection of these universals shows $\coprod\coprod z = \varphi$, and thus ψ is continuous, completing the proof. ∎

Lemma A.0.4. *With notation as in proposition A.0.3, if $(x,p) \sim_f (y,q)$, then $i(x,p) \sim i(y,q)$.*

Proof. Since $\sim_f$ is generated by the rule stated in remark A.0.2, it suffices to show the statement for the generating rule. Let $x = \alpha^*(y) = \beta^*(z)$, where $\alpha^*(y)$ is as given in lemma A.0.1, and β^* is a composite of faces. Then $i(y, \alpha_*(p)) = (y, \alpha_*(p))$ and $i(z, \beta_*(p)) = (z, \beta_*(p))$. We have

$$(z, \beta_*(p)) \sim (\beta^*(z), p) = (x, p) = (\alpha^*(y), p) \sim (y, \alpha_*(p)).$$

■

Remark A.0.5. One may be concerned that we assumed the simplex x was degenerate throughout *proposition* A.0.3, and never considered the case where x is nondegenerate. If x is nondegenerate, then $\alpha^* = 1$, which is clealy epi since it is an isomorphism, and thus the same proof holds. The same logic applies to all scenarios in which one might be concerned we did not consider a nondegenerate case.

References

[Fri23] Greg Friedman. An elementary illustrated introduction to simplicial sets, 2023. https://arxiv.org/abs/0809.4221.

[GJ99] Paul G. Goerss and John F. Jardine. *Simplicial Homotopy Theory*, volume 174 of *Progress in Mathematics*. Birkhäuser Basel, 1999.

[Hov99] Mark Hovey. *Model Categories*, volume 63 of *Mathematical Surveys and Monographs*. American Mathematical Society, 1999.

[Joy08a] André Joyal. Quasi-categories and kan complexes. Lecture notes, 2008.

[Joy08b] André Joyal. The theory of quasi-categories and its applications. Unpublished manuscript, 2008.

[Lei14] Tom Leinster. *Basic Category Theory*, volume 143 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, 2014. Available online at https://arxiv.org/pdf/1612.09375.

[Lur09] Jacob Lurie. *Higher Topos Theory*, volume 170 of *Annals of Mathematics Studies*. Princeton University Press, 2009. Available online at https://www.math.ias.edu/~lurie/papers/HTT.pdf.

[Lur11] Jacob Lurie. If i want to study jacob lurie's books "higher topoi theory", "derived ag", what prerequisites should i have? MathOverflow, 2011. URL:https://mathoverflow.net/q/74643.

[May67] J. Peter May. *Simplicial Objects in Algebraic Topology*. Chicago Lectures in Mathematics. University of Chicago Press, 1967.

[May99] J. Peter May. *A Concise Course in Algebraic Topology*. Chicago Lectures in Mathematics. University of Chicago Press, 1999. Revised version available at https://www.math.uchicago.edu/~may/CONCISE/ConciseRevised.pdf.

[ML98] Saunders Mac Lane. *Categories for the Working Mathematician*, volume 5 of *Graduate Texts in Mathematics*. Springer, second edition, 1998.

[Qui67] Daniel G. Quillen. *Homotopical Algebra*, volume 43 of *Lecture Notes in Mathematics*. Springer-Verlag, Berlin, Heidelberg, 1967. Unoffical revised and modernized version available at https://aareyanmanzoor.github.io/assets/books/homotopical-algebra.pdf.

[Rie11] Emily Riehl. A leisurely introduction to simplicial sets. Expository notes available online, 2011. URL: https://math.jhu.edu/~eriehl/ssets.pdf.

[Rie14] Emily Riehl. *Categorical Homotopy Theory*, volume 24 of *New Mathematical Monographs*. Cambridge University Press, 2014. Available online at https://emilyriehl.github.io/files/cathtpy.pdf.

[Rie16] Emily Riehl. *Category Theory in Context*. Aurora: Dover Modern Math Originals. Dover Publications, 2016. Available online at https://emilyriehl.github.io/files/context.pdf.